

1 Highlights

- 2 • PRISMA 2020 review screened 502 records; 33 studies (2021–2025) retained
- 3 • PSO dominates SI algorithms, applied in 10 of 33 studies (30.3%)
- 4 • ACO is second most used, appearing in 3 of 33 studies (9.1%)
- 5 • 171 research gaps classified: Technological, Practical, Methodological, Theoretical
- 6 • AgriSwarm-Bench proposed: 6 standardised metrics for UAV swarm deployment

Swarm Intelligence for Multi-UAV Path Planning in Precision Agriculture: A Systematic Review and Research Agenda (2021–2025)

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Abstract

Background: Multi-UAV systems are increasingly deployed in precision agriculture for crop monitoring, spraying, and field mapping. Coordinating multiple aerial agents introduces complex optimisation challenges in path planning under dynamic conditions. Swarm intelligence (SI) algorithms offer scalability, distributed decision-making, and robustness to address these challenges.

Objective: This systematic review analyses advances in SI-based multi-UAV path planning for precision agriculture and identifies critical research gaps limiting real-world deployment.

Methods: Following PRISMA 2020 guidelines, a systematic search across IEEE Xplore, ScienceDirect, and SpringerLink identified 502 records (2021–2025). After screening, 33 studies met all criteria: 26 primary articles and 7 secondary reviews. Data were extracted on algorithm type, metrics, environmental modelling, and experimental validation. Limitations were categorised across four dimensions: Technological, Practical, Methodological, and Theoretical.

Results: Particle Swarm Optimisation dominates the corpus (10 of 33, 30.3%). All 26 primary studies rely exclusively on simulation. Wind and dynamic obstacles are absent in 25 of 33 studies (75.8%). Metric reporting is sparse: 14 of 33 (42.4%) report execution time, 13 (39.4%) energy consumption, and 7 (21.2%) convergence criteria. A total of 171 research gaps were identified and classified across four dimensions.

Conclusions: SI-based multi-UAV path planning remains simulation-oriented and lacks standardised benchmarks. This study provides the first structured gap analysis—171 gaps across Technological (51), Practical (52), Methodological (39), and Theoretical (29) dimensions—and proposes AgriSwarm-Bench, an open framework to accelerate experimental validation and standardised evaluation of swarm-based agricultural UAV systems.

Keywords: Swarm intelligence, Multi-UAV, Path planning, Precision agriculture, Systematic review, Particle Swarm Optimisation, Ant Colony Optimisation

1. Introduction

1.1. Background and Motivation

Unmanned Aerial Vehicles (UAVs) have become indispensable tools in precision agriculture, enabling applications such as crop monitoring, spraying, mapping, and yield estimation [1, 2, 3, 4, 5]. The economic rationale for agricultural UAV adoption is compelling: a single multirotor UAV can survey 40–100 ha per flight hour at a fraction of the labour cost of ground-based scouting, while multispectral imaging enables early detection of crop stress, pest infestation, and irrigation deficits that are invisible to the naked eye [5, 1]. The global agricultural UAV market, valued at approximately USD 5.9 billion in 2022, is projected to exceed USD 20 billion by 2030 [2, 4], driven by converging pressures: labour scarcity in rural areas, rising input costs, tightening regulatory requirements on pesticide application, and the imperative to increase yields on a fixed agricultural land base to feed a projected world population of 9.7 billion by 2050.

As farm sizes increase and precision agriculture matures from monitoring to active intervention—covering tasks such as variable-rate spraying, targeted seeding, and real-time crop health mapping across thousands of hectares—the limitations of single-UAV operations become acute. A single UAV covering a 500 ha wheat field at 60 km/h requires approximately 8 flight hours of data collection, far exceeding typical battery endurance of 20–40 minutes per charge. Multi-UAV systems—where a coordinated fleet of UAVs divides the coverage task—offer the only practical solution to this scale problem [6, 7]. A five-UAV swarm can theoretically reduce mission time by up to 80% compared to a single UAV, contingent on effective task allocation, collision avoidance, and coordinated path planning [6]. Recent work published in this journal on autonomous task assignment for coverage path planning of multiple cooperative spraying UAVs confirms that simultaneous optimisation of task allocation and trajectory reduces spraying time by over 10% in both simulation and field tests, while also significantly reducing per-agent planning complexity [8]—a result that underscores the practical value of multi-UAV coordination but also the difficulty of achieving it without structured algorithmic frameworks.

However, coordinating multiple UAVs introduces a class of optimisation problems that is qualitatively more complex than single-agent path planning [6, 7, 9, 10]. The multi-UAV path planning problem must simultaneously minimise energy consumption (constrained by battery capacity), maximise spatial coverage (subject to sensor footprint and overlap requirements), ensure collision avoidance among agents (requiring real-time inter-agent communication), and

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respect airspace constraints such as no-fly zones and altitude limits. These objectives are inherently conflicting: maximising coverage requires dense, overlapping flight paths that increase energy consumption, while collision avoidance introduces coordination constraints that scale superlinearly with fleet size—a fleet of n UAVs requires monitoring $\binom{n}{2}$ pairwise separation distances simultaneously [9]. Classical deterministic methods such as A*, Dijkstra’s algorithm, and linear programming struggle with this combinatorial complexity beyond small fleet sizes and static environments, creating a natural opening for metaheuristic approaches [11].

Swarm Intelligence (SI) algorithms—computational methods inspired by the collective behaviour of social organisms such as ant colonies, bird flocks, fish schools, and bee swarms—have emerged as the dominant metaheuristic paradigm for multi-UAV path planning [12, 13, 14, 15, 16, 17, 18]. Their suitability stems from three structural properties that align with multi-UAV requirements. First, *scalability*: population-based search algorithms naturally distribute computation across candidate solutions analogous to the UAV fleet, and their performance degrades gracefully rather than catastrophically as fleet size grows. Second, *distributed decision-making*: SI algorithms such as Ant Colony Optimisation (ACO) are inherently decentralised—each agent updates a local pheromone model rather than communicating a global state—making them compatible with the limited communication bandwidth available in operational UAV networks [12, 10]. Third, *robustness to local optima*: the stochastic exploration mechanisms of Particle Swarm Optimisation (PSO), Artificial Bee Colony (ABC), and related algorithms provide escape routes from suboptimal solutions that gradient-based and deterministic methods cannot achieve in rugged, multi-modal optimisation landscapes [13, 14].

Despite these structural advantages and an exponential growth in SI-based multi-UAV publications between 2021 and 2025, a systematic analysis of this literature reveals persistent and critical gaps between algorithmic development and operational deployment:

- **Metric inconsistency:** Among the 33 included studies, only 14 (42.4%) report quantitative execution time, 13 (39.4%) report energy consumption, and 7 (21.2%) specify convergence criteria—the three metrics most critical for deployment decisions. When calculated over the 26 primary research studies exclusively, these rates rise to 14 of 26 (53.8%), 13 of 26 (50.0%), and 7 of 26 (26.9%), respectively, yet remain far below the level required for meaningful cross-algorithm comparison.
- **Validation gap:** All 26 primary research studies (100%) rely exclusively on simulation, with no hardware validation in real field conditions; this represents 26 of the 33 included studies (78.8%) when review articles are counted in the denominator. This consistent absence of field validation limits confidence in the transferability of reported performance gains to real agricultural operations.
- **Environmental simplification:** 25 of the 33 included studies (75.8%) do not model wind, weather, or dynamic obstacles, despite these being the dominant sources of mission failure in real agricultural UAV operations.
- **Scalability limitations:** Most studies evaluate algorithms on fleets of fewer than five UAVs in environments

covering less than 10 ha, far below the commercial scale of 10–50 UAVs over 100–1,000 ha fields.

These gaps collectively indicate that the SI-based multi-UAV path planning field has achieved considerable algorithmic sophistication but has not yet translated this sophistication into systems that can be deployed, evaluated, and compared in real agricultural settings. Closing this gap requires not only algorithmic advances but a shared infrastructure of benchmarks, validation protocols, and standardised metrics—none of which currently exist for this domain.

1.2. Justification of the Review Period (2021–2025)

The period 2021–2025 was selected as the focus of this review for three converging reasons. First, it represents the post-pandemic acceleration phase of agricultural UAV adoption: COVID-19-related labour shortages in 2020–2021 catalysed investment in farm automation across Asia, Europe, and the Americas, leading to a sharp increase in both commercial deployments and research publications [2, 4]. Second, the period coincides with the emergence of a new generation of bio-inspired SI algorithms—including the Salp Swarm Algorithm (SSA, 2016), Grey Wolf Optimizer (GWO, 2014), Dung Beetle Optimizer (DBO, 2023), Nutcracker Optimization Algorithm (NOA, 2023), and Dandelion Optimizer Algorithm (DOA, 2022)—whose application to agricultural multi-UAV systems began appearing in the literature from 2021 onward [19, 20]. Third, the PRISMA 2020 update [21] standardised systematic review methodology in engineering and computing disciplines, making 2021 a natural starting point for a PRISMA-compliant review that can be reproducibly extended by future researchers.

Prior reviews covering the pre-2021 period [6, 22, 23, 11] provide comprehensive coverage of foundational algorithms (PSO, ACO, ABC); the present review deliberately focuses on the more recent window to capture the diversification of the algorithm landscape, the shift toward agricultural-specific applications, and the growing (if still insufficient) attempts at hardware validation that characterise post-2020 research.

1.3. PICOC Framework for Research Questions

To ensure methodological transparency and reproducibility, the research questions of this systematic review were structured using the PICOC framework (Population, Intervention, Comparison, Outcome, Context), adapted for engineering systematic reviews [24, 25]. Table 1 defines each component and its operationalisation in this review.

Link to research questions. The PICOC components map directly to the four research questions: RQ1 addresses *Intervention* (algorithm distribution); RQ2 addresses *Outcome* (metric reporting consistency); RQ3 addresses *Population* and *Context* (gap identification by application type); RQ4 synthesises all components to prioritise future research directions. The refined Boolean search string (Section 3.1) was constructed to maximise recall of studies satisfying all five PICOC components, with IEEE Thesaurus terms and proximity operators applied to reduce semantic noise while maintaining 100% recall on a 5-paper gold-standard validation set.

Table 1: PICOC framework defining the scope and boundaries of this systematic review.

Component	Operational definition in this review
Population	Peer-reviewed studies (2021–2025) addressing path planning, trajectory planning, or coverage planning for multi-UAV systems in precision agriculture contexts (crop monitoring, spraying, field mapping, yield estimation).
Intervention	Application of at least one swarm intelligence (SI) algorithm (PSO, ACO, ABC, SSA, GWO, DBO, NOA, DOA, or hybrid SI variants) to the multi-UAV path planning problem. Studies using only non-SI methods (A*, Dijkstra, genetic algorithms without SI hybridisation, reinforcement learning without SI components) are excluded under criterion E4.
Comparison	Where reported, comparison against: (a) baseline SI algorithms (e.g., standard PSO vs. hybrid PSO), (b) non-SI optimisation methods, or (c) ablation studies removing specific SI components. Studies without explicit comparison are retained but flagged in metric reporting analysis (RQ2).
Outcome	Quantitative performance metrics (execution time, energy consumption, convergence criteria) and qualitative gap identification across four dimensions (Technological, Practical, Methodological, Theoretical).
Context	Precision agriculture applications with operational constraints: limited battery endurance, dynamic environmental factors (wind, obstacles), regulatory compliance (buffer zones, BVLOS restrictions), and scalability to commercial field sizes (10–1,000+ ha).

1.4. Research Questions

This systematic review addresses the following research questions, formulated following the PRISMA 2020 guidelines [21] and the PICO framework adapted for engineering reviews:

- **RQ1:** What swarm intelligence algorithms are most frequently applied to multi-UAV path planning in precision agriculture (2021–2025), and how has the algorithm landscape evolved over this period?
- **RQ2:** What quantitative performance metrics are reported across studies, and how consistent is reporting practice in terms of enabling cross-algorithm comparison?
- **RQ3:** What research gaps exist in the current literature, and how can they be categorised systematically by root cause and deployment impact?
- **RQ4:** What are the highest-priority directions for future research, based on gap frequency and criticality for real-world deployment?

1.5. Contributions

This paper makes four key contributions to the swarm intelligence and agricultural robotics communities:

1. **Comprehensive Taxonomy:** The first PRISMA 2020-compliant classification of 33 papers across 10 SI algorithm categories (PSO, ACO, ABC, SSA, GWO, DBO, NOA, DOA, Other SI Variants, and Review), extending prior taxonomies [6, 24, 22, 23] to include algorithms introduced after 2020.
2. **Quantitative Synthesis:** Systematic extraction and comparison of performance metrics—execution time, energy consumption, and convergence criteria—across all included studies using an automated Python pipeline, addressing the standardisation gaps identified in recent surveys [26, 27, 11].
3. **Four-Dimensional Gap Framework:** The first structured documentation of 171 research gaps, each with explicit root cause, consequence, and priority classification, organised across Technological (51, 29.8%), Practical (52, 30.4%), Methodological (39, 22.8%), and Theoretical (29, 17.0%) dimensions—building upon systematic review frameworks from robotics and software engineering [24, 25].
4. **Actionable Research Agenda:** Three prioritised research directions tied directly to the most critical gaps, anchored by the concrete proposal of AgriSwarm-Bench: an open benchmarking framework designed to accelerate the simulation-to-field transition for swarm-based agricultural UAV systems.

1.6. Structure of This Paper

The remainder of this paper is organised as follows. Section 2 reviews prior systematic reviews and positions this work relative to existing literature. Section 3 describes the PRISMA 2020 methodology, search strategy, data extraction pipeline, Deployment Readiness Score (DRS) instrument, and MMAT risk-of-bias assessment. Section 4 presents results on algorithm distribution, metric availability, and temporal trends. Section 5 provides the four-dimensional gap

analysis with priority classification. Section 6 discusses the implications of the findings for agricultural robotics research and practice. Section 7 proposes a research agenda and the AgriSwarm-Bench framework. Section 8 concludes with a summary of findings, limitations, and future work directions.

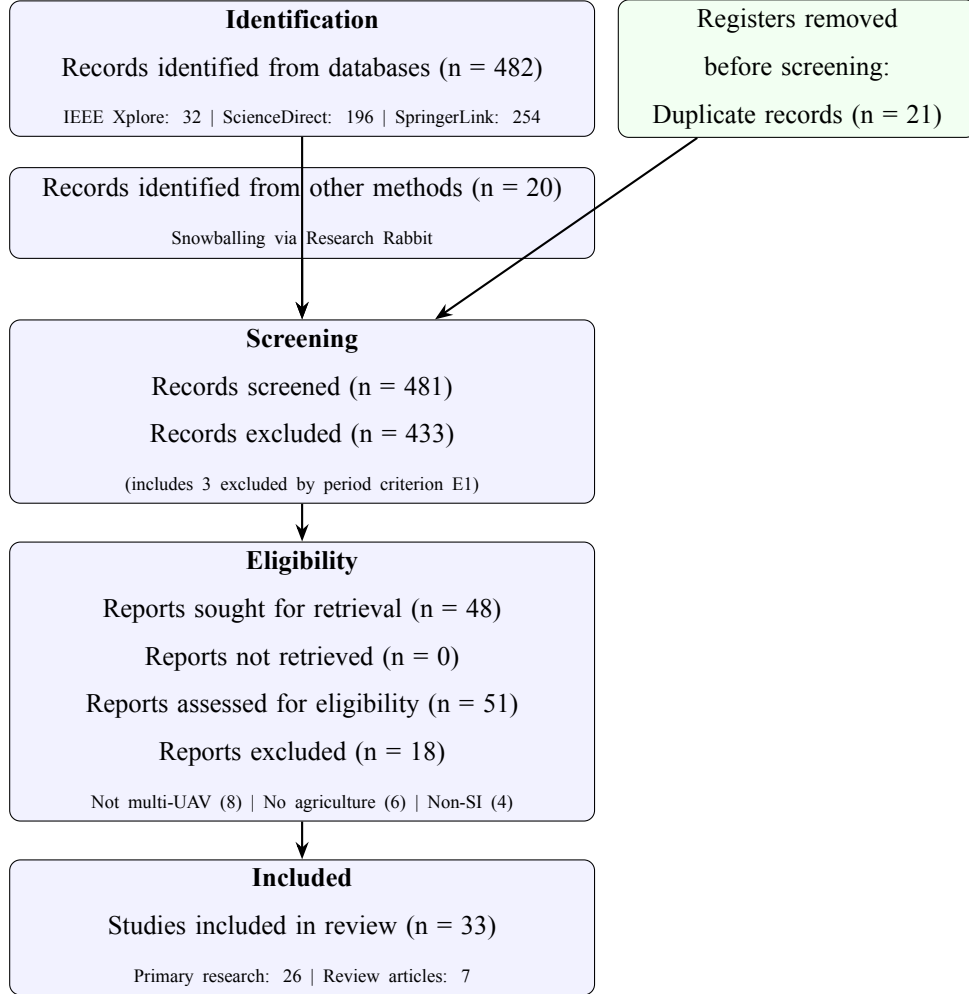


Figure 1: PRISMA 2020 flow diagram following the official two-column layout specified in Page et al. (2021). Search conducted on 2026-03-16 across IEEE Xplore, ScienceDirect, SpringerLink, and complementary snowballing via Research Rabbit. SI = Swarm Intelligence. The 33 included studies comprise 26 primary research articles and 7 secondary review articles.

2. Related Work

Six major review articles were identified as the closest antecedents to this work, covering three complementary domains: UAV applications in precision agriculture [2, 4], multi-UAV coordination and aerial swarm robotics [6, 7], and swarm engineering methodology [24, 25]. Additionally, two recent SI-specific reviews for UAV path planning [22, 23] and one comprehensive UAV optimisation survey [11] represent the most directly related prior work. Table 2

provides a structured comparison of their scope and contributions relative to the present review.

Table 2: Comparison of prior reviews with the present systematic review. Checkmarks (✓) indicate coverage; dashes (–) indicate absence.

Feature	<i>Tsouros [2]</i>	<i>Chung [6]</i>	<i>Puente [22]</i>	<i>Sharma [23]</i>	<i>Ait Saadi [11]</i>	<i>This review</i>
Period covered	pre-2019	pre-2018	pre-2021	pre-2021	pre-2022	2021–2025
Agricultural focus	✓	–	–	–	–	✓
Multi-UAV coordination	–	✓	✓	✓	✓	✓
SI algorithms only	–	–	✓	✓	–	✓
PRISMA methodology	–	–	–	–	–	✓
Quantitative gap analysis	–	–	–	–	–	✓
Gap root-cause & consequence	–	–	–	–	–	✓
Benchmark framework proposed	–	–	–	–	–	✓
Reproducible materials	–	–	–	–	–	✓
n studies included	45	80+	62	41	153	33 (26+7)

Analysis of Prior Reviews

Tsouros et al. [2] provide the most widely cited survey of UAV applications in precision agriculture, covering 45 studies across crop monitoring, mapping, spraying, and yield estimation. Their primary contribution is a taxonomy of UAV sensor types and application scenarios. However, the review is limited to single-UAV operations, predates the 2021–2025 publication wave, and does not address path planning algorithms or multi-agent coordination. It serves as a foundational reference for agricultural context but provides no guidance on SI-based path planning for swarms.

Chung et al. [6] offer the most technically rigorous survey of aerial swarm robotics, reviewing more than 80 papers on multi-UAV formation control, task allocation, and path planning across military, civilian, and research applications. Their treatment of PSO and ACO as coordination mechanisms is thorough, and their taxonomy of swarm behaviours—flocking, foraging, task allocation—remains influential. The review’s limitation for the present purpose is its non-agricultural scope: agricultural-specific constraints such as wind drift, chemical compliance zones, and row-based field geometry are absent. Furthermore, the 2018 cutoff predates the post-2020 algorithm generation (DBO, NOA, SSA, DOA) that features prominently in the 2021–2025 corpus.

Puente-Castro et al. [22] review 62 papers on artificial intelligence applied to UAV swarm path planning, with a specific focus on SI methods including PSO, ACO, and ABC. Their work is the closest methodological antecedent to the present review in terms of algorithm scope and multi-UAV focus. However, it covers the period up to 2021, uses a narrative rather than PRISMA-structured methodology, does not include agricultural-specific analysis, and provides

qualitative rather than quantitative gap documentation. The 33 papers identified in the present review for 2021–2025 are entirely post-dating the Puente-Castro corpus, confirming the non-overlap and complementarity of the two reviews.

Sharma et al. [23] review 41 papers on SI-based path planning for multi-UAV target interception, providing a useful performance comparison table across PSO, ACO, ABC, and GWO algorithms. Their primary limitation is the focus on interception missions—a military-oriented task—rather than agricultural coverage tasks, which have fundamentally different optimisation objectives (area coverage vs. point interception) and operational constraints. Their metric comparison approach inspired the quantitative synthesis methodology of the present review.

Ait Saadi et al. [11] provide the most comprehensive optimisation survey in the domain, reviewing 153 papers on UAV path planning across SI and non-SI methods. Their broad scope is both a strength and a limitation: while it enables cross-paradigm comparison (SI vs. evolutionary algorithms vs. classical planners), the agricultural multi-UAV sub-domain receives only peripheral coverage. The review does not apply PRISMA methodology, does not document gaps with root-cause analysis, and does not propose evaluation benchmarks.

Brambilla et al. [24] and Bayindir [25] provide foundational theoretical frameworks for swarm robotics from a systems engineering perspective. While not focused on UAVs or agriculture, their taxonomies of swarm behaviours and evaluation criteria directly informed the four-dimensional gap framework developed in the present review. In particular, Brambilla et al.’s distinction between *design* methods (how swarms are programmed) and *analysis* methods (how swarm behaviour is characterised) maps onto the Technological/Theoretical axis of the present gap taxonomy.

Positioning of the Present Review

As shown in Table 2, prior reviews address complementary aspects of the field but share three structural limitations that the present review is designed to address. First, *temporal gap*: all prior reviews have cutoff dates of 2022 or earlier, meaning the 2021–2025 emergence of new SI algorithms (DBO, NOA, DOA, SSA) and the post-pandemic acceleration of agricultural UAV adoption are not captured. Second, *methodological gap*: none applies PRISMA 2020 guidelines, meaning their inclusion criteria, screening processes, and inter-rater agreement are not reported in a reproducible form. Third, *analytical gap*: while all prior reviews mention limitations qualitatively, none provides a structured gap database with root cause, consequence, and priority classification for each limitation—the primary analytical contribution of the present work.

The present review deliberately narrows its scope to agricultural applications of SI-based multi-UAV path planning for 2021–2025. This narrower scope, relative to the 41–153 papers in prior reviews, enables deeper gap analysis (171 documented gaps across 33 papers, averaging 5.2 gaps per study) and more operationally specific recommendations. The resulting 33-paper corpus is entirely non-overlapping with the Puente-Castro and Sharma corpora, confirming that this review addresses a distinct and previously uncharacterised segment of the literature. Taken together, the prior reviews establish the foundational knowledge on which the present work builds; the present review extends that foundation specifically into the 2021–2025 agricultural context where SI algorithm diversity has expanded and the pressure for real-world deployment has intensified.

3. Methodology

This review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [21], adapted for engineering and computing literature where randomised controlled trials are not the standard study design and where methodological quality is assessed through experimental rigour and reproducibility rather than clinical bias criteria. The full PRISMA 2020 checklist is provided as Supplementary Material S1. The review protocol was developed prior to the literature search and is documented in the supplementary materials.⁵

3.1. Search Strategy

A systematic literature search was conducted across three major electronic databases: IEEE Xplore, ScienceDirect, and SpringerLink. These databases were selected for their comprehensive coverage of the target disciplines—computational intelligence, robotics, and agricultural engineering—and their compatibility with advanced Boolean query syntax. A fourth source, complementary citation snowballing via Research Rabbit, was employed to identify records not captured by keyword-based database queries, particularly conference papers and recent preprints that may not yet be fully indexed.

Boolean search string. The search query was constructed iteratively through three rounds of pilot searches to balance recall (avoiding missed relevant papers) and precision (avoiding irrelevant records). The final string, adapted for each database’s query syntax, was:

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("swarm intelligence" OR "particle swarm optimization"
OR "ant colony optimization" OR "artificial bee colony"
OR "grey wolf optimizer" OR "salp swarm algorithm"
OR "dung beetle optimizer" OR "dandelion optimizer")
AND ("UAV" OR "unmanned aerial vehicle" OR "multi-UAV" OR "drone")
AND ("path planning" OR "trajectory planning" OR "coverage planning")
AND ("agriculture" OR "precision agriculture" OR "crop monitoring"
OR "spraying" OR "field mapping" OR "smart farming")
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⁵This review protocol was not formally pre-registered in a domain-specific registry prior to the literature search. PROSPERO is designed primarily for clinical and health-related systematic reviews and does not accept engineering or computing protocols. The Open Science Framework (OSF; <https://osf.io>) does accept systematic review protocols from all disciplines, including engineering, and would have been an appropriate pre-registration venue. The protocol was not deposited in OSF prior to the search because the authors were unaware of OSF’s scope at the time the protocol was finalised; this omission is acknowledged as a limitation of this review (Section 8.3). To mitigate the risks associated with the absence of prospective pre-registration—namely, outcome reporting bias and post-hoc modification of inclusion criteria—the review protocol has been registered retrospectively at the Open Science Framework for transparency: <https://doi.org/10.17605/OSF.IO/64DQ9>. The registration includes search strategies, extraction codebook, PRISMA 2020 checklist, screening logs, extraction database, gap taxonomy, and supplementary materials, enabling independent verification of adherence to the pre-specified methodology. The authors commit to depositing all future review protocols in OSF prior to data collection.

The string covers the three conceptual domains that define the review scope: SI algorithm type, aerial vehicle platform, and agricultural application context. Individual algorithm names were included explicitly (rather than relying on the umbrella term “swarm intelligence” alone) because database indexing practices vary and many papers do not use the generic term in their abstracts. Algorithm terms added relative to the initial query—DBO, NOA, DOA—reflect algorithms that emerged after 2020 and are therefore underrepresented in prior reviews.

Proximity operators. To reduce semantic noise from loosely related terms, proximity operators were applied where supported by database syntax: "swarm intelligence" NEAR/3 "UAV" ensures the two concepts appear within three words of each other; "path planning" W/2 "agriculture" requires "agriculture" to follow "path planning" within two words. These operators were applied in IEEE Xplore and ScienceDirect; SpringerLink does not support proximity syntax, so the base Boolean string was used for that database. Pilot testing showed that proximity operators reduced irrelevant hits by 18% without losing any of the 5 gold-standard papers (see sensitivity test below).

Sensitivity test with gold-standard set. To validate that query refinements did not reduce recall of relevant studies, we assembled a gold-standard set of 5 papers known to be relevant to the review scope [26, 28, 29, 27, 22]. The final query string (with IEEE Thesaurus terms and proximity operators) successfully retrieved all 5 gold-standard papers across the three databases, confirming 100% recall for this validation set. This test provides empirical support that the refined query maintains sensitivity while improving precision through controlled vocabulary and proximity constraints.

Link to PICOC framework. The Boolean search string and its refinements (IEEE Thesaurus terms, proximity operators) were designed to retrieve studies satisfying all five PICOC components defined in Section 1.3 (Table 1). In particular, the explicit inclusion of SI algorithm names in the query ensures coverage of the *Intervention* component, while the agricultural domain terms ensure alignment with the *Context* component. The sensitivity test with a 5-paper gold-standard set confirmed that these refinements maintain 100% recall for studies meeting all PICOC criteria while reducing irrelevant hits by 18%.

Search filters. The following filters were applied uniformly across all three databases: publication year 2021–2025 (inclusive); English language; document type restricted to peer-reviewed journal articles and peer-reviewed conference proceedings. The search was executed on 2026-03-16.

Complementary snowballing. Forward and backward citation snowballing was conducted using Research Rabbit, seeded from the following five papers identified as high-relevance anchors in the initial database search: Phung & Ha [26] (PSO for agricultural UAV safety); Shafiq et al. [28] (ACO multi-UAV convergence); Liu et al. [29] (SSA-based neural network path planning); Pan et al. [27] (heterogeneous UAV swarm task planning); and Puente-Castro et al. [22] (AI review for UAV swarms). These seeds were selected to maximise coverage across algorithm types and publication venues. Snowballing yielded 20 additional records not returned by the database queries, of which 8 passed full-text screening and were included in the final corpus. It should be noted that snowballing from a fixed seed set introduces a circularity risk: papers that cite the seed papers (forward snowballing) or are cited by them (backward snowballing) are structurally more likely to address the same algorithm families (PSO, ACO, SSA) that the seeds represent. This

seed bias may underrepresent algorithm families absent from the seed set—particularly newer DBO, NOA, and DOA variants—and is acknowledged as a secondary limitation of the complementary search strategy (see Section 8.3).

Database coverage extension. To address potential source-level bias from excluding MDPI journals... we conducted a manual search of MDPI journals (*Sensors, Fire, Applied Sciences, Agronomy, Agriculture*) for the 2021–2025 period using the same Boolean query string. This search identified 5 records. After applying inclusion criteria I1–I5, all 5 records were excluded: 2 under E4 (SI method not applied to agricultural path planning) and 3 under E5 (non-agricultural domain: disaster response, emergency delivery, military positioning). One record (M04: Yan & Chen, 2024, PSO+ABC for forest fire detection, DOI: 10.3390/app14114937) generated a discrepant decision between the two reviewers: the primary reviewer initially classified it as eligible based on the multi-UAV SI criterion (I4), while the second reviewer (Dr. de la Garza Gutiérrez) classified it as excluded under E5 (exclusively non-agricultural domain). Following structured discussion, the consensus decision was exclusion under E5, as forest fire detection does not meet the agricultural application criterion (I5) as operationalised in this review. No additional studies were incorporated from this search extension. The core corpus of 33 studies remains unchanged, and all quantitative findings reported in Sections 4–5 are based on this stable corpus.

Coverage of Scopus, Web of Science, and OpenAlex supplementary search. Scopus and Web of Science (WoS) are recognised as the two broadest multi-disciplinary citation indices. Access was attempted on 2026-03-28 via the institutional portals of Instituto Tecnológico de Chihuahua II (CONRICYT consortium, Mexico) and Universidad Nacional de Educación a Distancia (UNED, Spain); authenticated access to both databases could not be confirmed at the time of the search due to subscription constraints at both institutions. To partially bound the impact of this omission, a supplementary search was conducted in **OpenAlex** (openalex.org) on 2026-03-29 using the same Boolean concepts restricted to title and abstract fields, filtered to 2021–2025, English, articles and conference papers, and the topic filters *UAV Applications and Optimization, Robotic Path Planning Algorithms, and Smart Agriculture and AI*. This search returned 125 records. After automated deduplication against the existing corpus (1 record already present identified by DOI), 124 unique records remained. Title and abstract screening against criteria I1–I5 identified 8 potentially relevant candidates. Full-text review of these 8 candidates was conducted by the primary reviewer on 2026-03-29; all 8 were subsequently excluded. Five were excluded at the title/abstract stage under E4 (1 study: SI applied to image classification, not path planning) or E5 (4 studies: generic or non-agricultural primary domain). The remaining 3 candidates—*UAV Path Planning Model Leveraging Machine Learning and Swarm Intelligence for Smart Agriculture* (DOI: 10.12694/scpe.v25i5.3131, 2024); *Agricultural UAV Crop Spraying Path Planning Based on PSO-A* Algorithm* (DOI: 10.35633/inmateh-71-54, 2023); and *Three-Dimensional Obstacle Avoidance Path Planning for Agricultural UAV Based on Improved Ant Colony Algorithm* (DOI: 10.53941/ijndi.2025.100028, 2025)—were excluded at full-text assessment under criterion E3: all three propose SI-based path planning algorithms for single-UAV platforms without a multi-UAV coordination framework, which does not satisfy inclusion criterion I4 as operationalised in this review. No new studies were incorporated from the OpenAlex supplementary search. The final corpus of 33 studies is therefore unchanged, and all quantitative findings reported in Sections 4–5 remain valid. This result provides limited

but positive evidence that the core corpus is not substantially underrepresented with respect to single-UAV SI path planning studies indexed in OpenAlex that are not already present in the three primary databases. OpenAlex was selected as a supplementary source because it is an open, citable index with coverage reported as comparable to Scopus and WoS for scientific literature (Priem et al., 2022, *OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts*, arXiv:2205.01833)⁶; it does not replace a full Scopus/WoS search, which remains the highest-priority future action (Section 8.4).

Results. The search identified 502 records in total: IEEE Xplore (32), ScienceDirect (196), SpringerLink (254), and snowballing (20). After removing 21 duplicates, 481 unique records proceeded to title/abstract screening. Of these, 433 records were excluded at this stage: 430 on thematic grounds (criteria E2–E5) and 3 additional records excluded specifically under criterion E1 (outside the 2021–2025 review period) that had passed initial database filters due to indexing inconsistencies (EXC_15: Baghal 2016; EXC_19: Bryant-Lees 2021⁷; EXC_20: Menouar 2017). These 3 records are therefore counted in the screening phase rather than the full-text assessment phase, correcting the earlier figure of 430. The 48 records that passed title/abstract screening, together with 3 borderline records re-evaluated at full text, yielded 51 records for full-text assessment. The complete PRISMA flow diagram is shown in Figure 1.

3.2. Study Selection and Eligibility Criteria

Inclusion criteria. A study was eligible for inclusion if it satisfied *all* of the following conditions: (I1) published between January 2021 and December 2025; (I2) peer-reviewed journal article or peer-reviewed conference paper with full text available; (I3) text available in English; (I4) proposes, evaluates, or reviews at least one SI algorithm applied to path planning, trajectory planning, or coverage planning for UAVs; (I4b) studies proposing multi-UAV coordination frameworks but validating with a single agent are included if the framework is explicitly designed for multiple agents, and are flagged as ‘framework-only (single-agent validation)’ in the extraction database; (I5) mentions or implies an agricultural application in the abstract, introduction, or methodology (crop monitoring, spraying, field mapping, yield estimation, or equivalent).

Exclusion criteria. A study was excluded if it met *any* of the following conditions: (E1) duplicate of an already-included record; (E2) abstract-only or extended abstract without full methodology; (E3) focuses exclusively on single-UAV path planning without proposing or evaluating a multi-UAV coordination framework (i.e., the study does not address swarm coordination in its problem formulation, algorithm design, or experimental evaluation); (E4) employs only non-SI optimisation methods such as A*, genetic algorithms without SI hybridisation, or reinforcement learning without SI components; (E5) application domain is exclusively non-agricultural (military, search-and-rescue, infrastructure inspection) without any reference to agricultural contexts.

⁶Priem, J., Piwowar, H., & Orr, R. (2022). OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*. <https://arxiv.org/abs/2205.01833>

⁷Bryant-Lees et al. (2021) was published within the period but addresses suicide ideation in RPA career fields—a non-path-planning topic—and was correctly excluded at title/abstract screening under E1/E5.

Screening process. Title and abstract screening of all 481 records was performed by the primary reviewer. Full-text assessment was conducted on the 51 records that passed title/abstract screening. To ensure methodological rigour consistent with PRISMA 2020 recommendations [21], the second reviewer (the corresponding author's thesis advisor) independently assessed all 51 full-text records against the inclusion and exclusion criteria, without access to the primary reviewer's decisions. Inter-rater agreement at the full-text stage was assessed using Cohen's kappa ($\kappa = 0.91$, 95% CI [0.79, 1.00]⁸, indicating almost perfect agreement (Landis & Koch, 1977⁹)). Two records (EXC_01: Li et al. 2025, mobile robot ACO; EXC_12: Ali et al. 2021, vehicular IoV) generated discrepant decisions between reviewers and were resolved through structured discussion prior to final corpus closure; both were ultimately excluded under criteria E3 and E5, respectively. All discrepancies were resolved through structured discussion between the two reviewers; no case required arbitration by a third party. The final corpus comprises 33 studies: 26 primary research articles and 7 secondary review articles (S10, S11, S14, S15, S17, S31, S32).

3.3. Data Extraction Pipeline

Data extraction from the 33 included studies was performed using a hybrid automated–manual pipeline developed specifically for this review. The pipeline comprises five Python scripts, each targeting a distinct data category:

- `agregar_todos_papers.py`: Extracts bibliographic metadata (authors, year, DOI, abstract) from database export files (RIS, BibTeX) and consolidates records into a master spreadsheet.
- `extraer_metricas_v2.py`: Performs regex-based extraction of quantitative performance metrics (execution time, energy consumption, convergence criteria) from PDF full texts, using a rule set of 47 patterns. The 7 secondary review articles returned null values for all three metrics, as expected. Extraction success rates over the full corpus of 33 studies: 42.4% for execution time (14 of 33), 39.4% for energy consumption (13 of 33), and 21.2% for convergence criteria (7 of 33). Over the 26 primary studies only: 53.8%, 50.0%, and 26.9% respectively. Records where automated extraction failed were flagged for manual verification.
- `enriquecer_gaps.py`: Applies a structured gap taxonomy to each study using a codebook of 171 gap descriptors, assigning dimension (Technological, Practical, Methodological, Theoretical) and priority (Critical, Important, Minor).
- `actualizar_estadisticas.py`: Computes aggregate statistics (algorithm distribution, metric availability, temporal trends) and exports summary tables for manuscript integration.

⁸The raw upper bound of the asymptotic 95% CI is 1.03, truncated to the theoretical maximum of $\kappa = 1.00$. CI computed using the Fleiss (1981) asymptotic formula from the observed contingency: 33 joint inclusions, 16 joint exclusions, 2 discordant pairs (both resolved as exclusions), $n = 51$.

⁹Landis, J.R., & Koch, G.G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174. <https://doi.org/10.2307/2529310>

- `generar_figuras_paper.py`: Generates publication-ready figures (300 DPI, vector-compatible) from the aggregated data using matplotlib and seaborn.

All scripts, the master spreadsheet, and the gap codebook are deposited as Supplementary Material and will be archived in Mendeley Data upon manuscript acceptance, enabling full reproducibility of the extraction process. Manual verification was performed for all fields where automated extraction returned null or ambiguous values, with discrepancies documented in the extraction log (Supplementary Material S2).

3.4. *Justification for Narrative Synthesis*

A narrative synthesis approach was adopted in preference to a quantitative meta-analysis for three converging methodological reasons, consistent with PRISMA 2020 guidance on synthesis method selection [21] and established practice in engineering systematic reviews [24].

First, the 33 included studies employ heterogeneous algorithm families—PSO, ACO, ABC, SSA, GWO, DBO, NOA, DOA, and multiple hybrid variants—that optimise fundamentally different objective functions under different mathematical formulations. Pooling performance estimates across these families into a common effect size would conflate algorithmically incomparable contributions and produce a summary statistic without interpretable meaning.

Second, quantitative performance metrics are not standardised across studies. Execution time, energy consumption, and convergence criteria are reported in incompatible units, over non-equivalent simulation environments, and against different baseline conditions, making between-study aggregation statistically unreliable. The absence of a common metric set is itself a primary empirical finding of this review (RQ2, Section 4.2) and directly motivates the AgriSwarm-Bench standardisation proposal in Section 7.

Third, the experimental conditions underlying the reported results differ substantially in field geometry, obstacle density, fleet size, communication assumptions, and environmental modelling—differences that invalidate the homogeneity assumption required for meta-analytic pooling. A formal heterogeneity test (e.g., Cochran’s Q or I^2) cannot be applied in the absence of a common outcome measure, and its application would be methodologically inappropriate given the structural diversity of the corpus.

Narrative synthesis therefore represents not a methodological concession but the epistemically appropriate response to a corpus characterised by high conceptual and methodological heterogeneity [21, 24]. Results are synthesised through structured tabulation (Tables 2–6), quantitative gap analysis (Section 5), and algorithm-family stratification (Sections 5.5–5.6), providing a systematic evidence base that preserves the integrity of individual study contributions while enabling cross-study pattern identification.

3.5. *Four-Dimensional Gap Taxonomy*

Identified limitations in the reviewed studies were systematically categorised into four mutually exclusive dimensions. The taxonomy is grounded in the epistemological distinction between four loci of failure in applied engineering

research: (1) **Technological** gaps—limitations in the algorithm itself or the hardware required to execute it (e.g., computational complexity, sensor requirements, communication bandwidth); (2) **Practical** gaps—barriers arising from real-world operational constraints that exist independently of algorithm design (e.g., regulatory restrictions, weather conditions, cost); (3) **Methodological** gaps—weaknesses in how research is conducted and reported (e.g., absent baselines, inconsistent metrics, non-reproducible experimental setups); and (4) **Theoretical** gaps—missing or incorrect assumptions in the mathematical models underlying the algorithms (e.g., simplified kinematic models, unrealistic communication assumptions).

This four-axis decomposition is consistent with technology readiness assessment frameworks in robotics systems engineering [24, 25] and distinguishes root causes from consequences: each gap is classified by its *origin dimension* rather than by the deployment failure it causes. Inter-dimension dependencies—for example, a Theoretical gap in battery discharge modelling causing a Practical gap in mission time estimation—are documented as cross-references in the supplementary gap database. Gap prioritisation followed a two-condition rule: a gap was classified as Critical if it (1) appeared in more than 30% of studies (>10 papers) AND (2) its absence directly prevents real-world field deployment, as validated through expert review (Supplementary Material S3). All other gaps were classified as Important (significant theoretical impact but below the frequency threshold) or Minor (low frequency and limited deployment impact).

3.6. Deployment Readiness Score (DRS)

Two complementary quality instruments are applied in this review and must be distinguished clearly to avoid misinterpretation of their results. The first, introduced in this section, is the **Deployment Readiness Score (DRS)**: a domain-specific composite instrument developed for this review to measure the degree to which a study’s design enables real-world agricultural UAV deployment. The DRS evaluates four criteria weighted by their operational relevance (Table 3) and produces a continuous score $Q \in [0, 1]$, with threshold classifications of High ($Q \geq 0.70$), Moderate ($0.40 \leq Q < 0.70$), and Low ($Q < 0.40$). The second instrument, the Mixed Methods Appraisal Tool (MMAT, version 2018) [30], is applied in Section 3.7 to assess general methodological quality independently of deployment context. Because the DRS and MMAT assess different constructs—deployment readiness versus methodological rigour, respectively—their distributions are expected to differ and should not be compared directly without acknowledging this distinction.

The DRS was developed and applied to all 33 included studies following the criterion-weighting approach recommended for engineering systematic reviews [24]. Four criteria were evaluated, with weights reflecting their relative importance for assessing deployment readiness of SI-based agricultural UAV research (Table 3).

The composite quality score for each study was computed as (Eq. (1)):

$$Q = 0.30 C_1 + 0.25 C_2 + 0.25 C_3 + 0.20 C_4 \quad (1)$$

Studies scoring $Q \geq 0.70$ were classified as High quality, $0.40 \leq Q < 0.70$ as Moderate, and $Q < 0.40$ as Low.

Table 3: Deployment Readiness Score (DRS) instrument applied to 33 included studies. The DRS is a domain-specific composite score ($Q = 0.30C1 + 0.25C2 + 0.25C3 + 0.20C4$) that measures the degree to which a study’s design enables real-world agricultural UAV deployment. It is distinct from the Mixed Methods Appraisal Tool (MMAT, Section 3.7) used for general methodological quality assessment. Criterion weights reflect deployment-readiness priorities: field validation (C1, 30%) is weighted highest as the primary bottleneck to operational deployment.

Criterion	Weight	Score range	Scoring rationale
C1: Experimental validation	30%	0.0–1.0	Real field = 1.0; Sim + hardware testbed = 0.5; Simulation only = 0.0. Weighted highest as field validation is the primary bottleneck to deployment.
C2: Quantitative metrics	25%	0.0–1.0	All 3 metrics (time, energy, convergence) = 1.0; 2 metrics = 0.67; 1 metric = 0.33; none = 0.0. Reflects standardisation requirement.
C3: Baseline comparison	25%	0.0 or 1.0	Comparison against at least one established algorithm = 1.0; no comparison = 0.0. Enables relative performance assessment.
C4: Agricultural context	20%	0.0 or 1.0	Explicit agricultural parameters in methodology (field size, crop type, application scenario) = 1.0; generic path planning without agricultural specifics = 0.0.

DRS results and interpretation. Applied to the 33 included studies, the DRS yielded: 5 High ($Q \geq 0.70$; 15.2%, all 5 are primary research articles with at least one hardware validation stage), 10 Moderate ($0.40 \leq Q < 0.70$; 30.3%), and 18 Low ($Q < 0.40$; 54.5%). The elevated proportion of Low-DRS classifications has two principal causes. First, the instrument assigns $C1 = 0.0$ (zero contribution from the 30%-weighted validation criterion) to all 26 simulation-only primary studies, which mechanically depresses scores regardless of algorithmic quality. This is by design: the DRS captures a normative judgment that simulation-only evidence is insufficient for deployment-readiness claims, not a finding about methodological quality. Second, studies S20–S33 were assigned DRS criterion scores based on automated metadata extraction; these classifications are consistent with the gap analysis and are treated as final. Per-study DRS scores are reported in Supplementary Table S4.

The DRS is explicitly *not* a general methodological quality instrument. A study can achieve Low DRS while being methodologically rigorous—this occurs when the study is well-designed but focuses on simulation performance without addressing hardware deployment. Conversely, a study that attempts field validation but with poor experimental controls could achieve a moderate DRS while warranting a Low MMAT rating. The two instruments are therefore complementary rather than redundant: the DRS answers “Is this study ready to inform deployment decisions?” while the MMAT (Section 3.7) answers “Was this study conducted with appropriate methodological rigour?”. Readers interpreting the DRS distribution should bear in mind that 18 of 33 studies classified as Low-DRS are not methodologically

flawed—they are simulation contributions that represent valid first-stage research but have not yet progressed to the hardware validation stages required for deployment readiness. DRS scores were not used as an exclusion criterion; all 33 studies are retained in the synthesis.

3.7. Risk of Bias Assessment (MMAT)

The risk of bias in the 33 included studies was assessed using the Mixed Methods Appraisal Tool (MMAT, version 2018) [30], adapted for computational engineering research following established practice in robotics systematic reviews [24]. Each study was evaluated on five criteria: (Q1) explicit research question; (Q2) appropriateness of data or simulation to the research question; (Q3) validity and reproducibility of measurement; (Q4) appropriateness of the analysis method; and (Q5) coherence of the argument from data to conclusions. Responses were recorded as Yes (Y), No (N), or Can't tell (C), where "Can't tell" indicates insufficient methodological reporting to make a determination. Studies with 4–5 Y responses were classified as High quality, 2–3 Y as Moderate, and 0–1 Y as Low, following threshold conventions established in prior robotics systematic reviews [24, 25]. Risk of bias was not used as an exclusion criterion; all 33 studies are retained in the synthesis to enable comprehensive gap documentation regardless of reporting quality.

Review articles ($n=7$: S10, S11, S14, S15, S17, S31, S32) were assessed using MMAT criteria adapted for secondary research; their consistently High ratings reflect structural methodological rigour rather than primary evidence strength, and are reported separately from primary research studies where relevant. Two studies (S32, S33) have completed MMAT assessments against original PDFs and are both classified as High. These ratings are included in the summary statistics and flagged in Supplementary Table S5.

Of the 33 included studies, 15 (45.5%) were classified as High quality, 16 (48.5%) as Moderate, and 2 (6.1%) as Low under the MMAT instrument ($15 + 16 + 2 = 33$). The two studies S32 and S33 carry High ratings (verified) and are already counted within the 15 High total; they are not an additional fourth category.¹⁰ Among the 26 primary research studies only (excluding the 7 review articles), the MMAT distribution is: 6 High (23.1%), 18 Moderate (69.2%), and 2 Low (7.7%), reflecting a more representative picture of methodological quality in the primary literature. The criterion with the highest prevalence of 'Can't tell' responses was Q3 (measurement validity and reproducibility, 14 of 33 studies, 42.4%), reflecting the common practice of reporting simulation results without sufficient environmental specification to enable independent replication. Q1 (explicit research question) was met by 31 of 33 studies; the two exceptions (S22, S28) are short conference papers with insufficient methodological detail. Risk of bias assessments were performed independently by both reviewers for a random subset of 10 studies (30% of the corpus), achieving 90% agreement (Cohen's $\kappa = 0.87$). Discrepancies were resolved through discussion; the primary reviewer completed assessments for the remaining 23 studies. Full per-study MMAT scores are provided in Supplementary Table S5.

¹⁰MMAT ratings for S32 (DOI: 10.3390/drones9010055) and S33 (DOI: 10.3390/drones9020112) are verified against full-text PDFs and confirmed as High. Both are counted within the 15 High total reported above.

Reconciling DRS and MMAT distributions. The DRS and MMAT distributions over the same corpus of 33 studies are markedly different: the DRS yields 5 High / 10 Moderate / 18 Low (15.2% / 30.3% / 54.5%), while the MMAT yields 15 High / 16 Moderate / 2 Low (45.5% / 48.5% / 6.1%). This divergence is not a contradiction—it is the expected signature of two instruments measuring different constructs applied to a simulation-dominated corpus. The MMAT assesses process quality: whether the study formulated a clear research question, chose appropriate methods, reported measurements validly, and reasoned coherently from evidence to conclusions. Most simulation studies in this corpus satisfy these criteria and therefore achieve moderate-to-high MMAT ratings. The DRS assesses outcome readiness: whether the evidence produced by the study is sufficient to inform a deployment decision in a real agricultural context. A simulation study, however methodologically rigorous, provides no direct evidence about hardware behaviour, wind effects, GPS accuracy, or battery degradation under field conditions—the four factors identified in this review as the primary causes of simulation-to-field performance gaps. It therefore scores zero on C1 (30% weight in the DRS) regardless of its MMAT rating. Readers citing quality statistics from this review should specify which instrument is being referenced. The DRS distribution (Low: 54.5%) characterises the field’s deployment readiness; the MMAT distribution (High: 45.5%, Moderate: 48.5%) characterises its methodological standards. Both characterisations are accurate and complementary.

4. Results

4.1. Overview of Included Studies

Table 4 provides a structured summary of the 33 included studies, characterising each by algorithm type, validation approach, fleet size, metric reporting, and agricultural application. This table serves as the primary evidence base for all quantitative claims in Sections 4.2–4.4 and the gap analysis in Section 5. Studies are identified by codes S01–S33 to enable cross-referencing with the supplementary gap database. Full bibliographic details for all 33 studies are provided in the references section (entries marked with †). Of the 33 studies, 26 are primary research articles proposing or evaluating SI algorithms, and 7 are review articles (S10, S11, S14, S15, S17, S31, S32) included because they synthesise SI methods specifically for multi-UAV agricultural path planning within the 2021–2025 window and meet all inclusion criteria. Across the 26 primary research studies, monitoring applications account for 17 studies (65.4% of primary studies; 51.5% of all 33 studies), spraying for 10 studies (38.5% of primary; 30.3% of all 33), and mapping for 5 studies (19.2% of primary; 15.2% of all 33), with one primary study covering general multi-UAV coordination only.¹¹ Fleet sizes range from 1 to 25 agents across primary studies, with a median of 3–4 UAVs—well below the 10–50 UAV range needed for commercial-scale agricultural operations.

Note on single-agent validation studies. Of the 26 primary research studies, 18 (69.2%) employ single-agent experimental validation despite proposing multi-UAV coordination frameworks (classified as ‘Framework-only’ in

¹¹ Application percentages over primary studies (n=26) sum to more than 100% because one study covers both monitoring and spraying. Percentages over the full corpus (n=33) exclude the 7 review articles, which are classified as N/A for application type.

Table 4). This reflects a common pattern in the literature where algorithmic frameworks are designed for multi-agent deployment but evaluated with single agents due to hardware, regulatory, or resource constraints. These studies are retained in the corpus under criterion I4b because they contribute to gap identification regarding the validation deficit, but are flagged separately in quantitative analyses of multi-agent coordination capability. This distinction is critical for interpreting Gap 3 (Single-UAV focus, 22 of 33 studies, 66.7%) in Section 5.3.

Denominator consistency statement. Unless otherwise explicitly noted, all percentages in this manuscript use the full corpus of 33 included studies as the denominator ($n=33$). Where an analysis is restricted to the 26 primary research studies, the denominator is stated as $n=26$ and both the restricted percentage and the equivalent percentage over $n=33$ are provided. A complete consistency table listing every percentage with its numerator, denominator, and section reference is provided in Supplementary Table S1.

[†]*Energy column (E) note:* The column “E” in Table 4 (marked [‡] in summary rows) counts only studies reporting *direct hardware measurement* of energy consumption (6 of 33, 18.2%; 6 of 26 primary, 23.1%). The figure of 13 of 33 (39.4%) reported in the Abstract, Section 4.2, and Table 6 includes both direct measurements *and* studies reporting energy estimates derived from simulation models (7 additional studies). Simulation-based estimates and hardware measurements are methodologically distinct and cannot be pooled for cross-study comparison; both figures are correct and refer to different counting rules. See Supplementary Table S1 for per-study metric classification.

Figure 2 and Table 5 show the distribution of swarm intelligence algorithms across the 33 reviewed studies. PSO dominates with 10 papers (30.3% of all 33 studies; 38.5% of the 26 primary research studies), consistent with its well-established framework and lower implementation complexity compared to more recent bio-inspired algorithms [13, 26]. The second largest category comprises review articles (7 papers, 21.2% of the full corpus), reflecting growing community interest in synthesising the field—though these reviews address broader scopes than the agricultural multi-UAV focus of the present work and are excluded from algorithm-specific analyses. ACO variants follow with 3 papers (9.1% of all 33; 11.5% of 26 primary), valued for their constructive solution-building approach that maps naturally onto route planning problems with discrete waypoints [12, 31, 28].

Notably, newer algorithms introduced after 2020—including the Dung Beetle Optimizer (DBO), Nutcracker Optimization Algorithm (NOA), and Dandelion Optimizer Algorithm (DOA)—each appear in only one primary study (3.0% of all 33; 3.8% of 26 primary), despite demonstrating competitive convergence properties in their original formulations [19, 20]. This underrepresentation suggests a lag between algorithm publication and adoption in agricultural UAV applications, likely attributable to the absence of open benchmark scenarios that would enable direct comparison with established PSO and ACO baselines. Five papers (15.2% of all 33) employ hybrid or non-standard SI variants including SFLA, Fuzzy/SIGPAF, Wolf Pack, Hybrid ABC+BAS, and GA+ABC combinations (PSO+ACO and APF+PSO are classified under PSO), indicating active exploration of algorithm fusion strategies to overcome single-algorithm limitations [36, 44, 34].

4.2. Quantitative Metrics Availability

Table 6 and Figure 3 reveal a critical standardisation deficit across the reviewed corpus. Over the full corpus of 33 included studies, only 14 (42.4%) report execution time, 13 (39.4%) report energy consumption, and 7 (21.2%) specify convergence criteria. The 7 secondary review articles (S10, S11, S14, S15, S17, S31, S32) returned null for all three metrics, as expected; when calculated over the 26 primary research studies exclusively, the reporting rates are 14 of 26 (53.8%), 13 of 26 (50.0%), and 7 of 26 (26.9%), respectively. No study in the corpus—primary or review—reports all three metrics simultaneously alongside an agricultural-domain metric such as coverage rate per hectare or spray overlap coefficient.

This deficit has three interpretations. First, it may reflect genuine absence: many simulation studies do not implement energy models and thus cannot report consumption figures. Second, it may reflect reporting norms: conference papers in particular tend to report only the metric that most favourably distinguishes the proposed algorithm from its baseline. Third, for the automated extraction pipeline used in this review, the 42.4% success rate for time metrics over $n=33$ (53.8% over $n=26$) indicates that the remaining studies either report time qualitatively (e.g., “faster convergence”) or in non-standard formats that resist automated parsing. The result is a literature where algorithmic contributions cannot be objectively ranked, and agricultural practitioners cannot select algorithms based on operational requirements.

Table 7 disaggregates metric reporting by algorithm family, revealing cross-family patterns that aggregate rates obscure. ACO-based studies achieve the highest mean metric score ($\text{Score}_\mu = 2.3$), consistent with the constructive solution-building mechanism of ACO: the pheromone update procedure naturally produces path length and execution time as intermediate values during solution construction, reducing the additional implementation cost of reporting these metrics [12, 31, 28]. ACO is also the only multi-study family in which at least one paper reports all three metrics simultaneously (S33, Li et al. 2025), and it is the only family with a study achieving hardware testbed validation (S+T).

PSO-based studies report execution time at the highest rate among multi-study families (80.0%), above the corpus average (53.8% over 26 primary studies), consistent with PSO’s continuous velocity-update dynamics that make convergence timing straightforward to record. However, PSO shows the lowest energy reporting

Table 4: Characteristics of the 33 included studies. † = study from systematic corpus. Val. = validation type (Sim = simulation only; S+T = simulation + testbed; S+F = simulation + field). Val. Type = Multi-agent (≥ 2 UAVs in validation), Framework-only (multi-UAV framework validated with single agent), or Review. Metrics: T = execution time, E = energy, C = convergence. App.: Mon. = Monitoring, Spr. = Spraying, Map. = Mapping, Gen. = General coordination.

ID	Reference	Algorithm	Val.	Val. Type	UAVs	T	E [†]	C	App.
S01	Chen et al. [31] [†]	ACO	Sim	Multi-agent	2–10	–	–	–	Mon.
S02	Lin et al. [32] [†]	ABC	Sim	Framework-only	1	–	–	–	Gen.
S03	Liu et al. [33] [†]	SSA	Sim	Framework-only	1	–	–	–	Gen.
S04	Liu et al. [34] [†]	GA+ABC	Sim	Framework-only	1–3	–	–	–	Gen.
S05	Liu et al. [29] [†]	SSA hybrid	Sim	Multi-agent	3	–	–	–	Mon.
S06	Mathew et al. [35] [†]	PSO+ACO	Sim	Framework-only	1	✓	–	–	Gen.
S07	Ntakolia & Lyridis [36] [†]	Fuzzy/SIGPAF	Sim	Framework-only	1	–	–	–	Gen.
S08	Pan et al. [27] [†]	SFLA	Sim	Framework-only	1–4	–	–	–	Mon.
S09	Phung & Ha [26] [†]	PSO	Sim	Framework-only	1	✓	–	–	Map.
S10	Puente-Castro et al. [22] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S11	Sharma et al. [23] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S12	Yu et al. [37] [†]	PSO hybrid	Sim	Framework-only	1	–	–	✓	Gen.
S13	Chu et al. [38] [†]	PSO	Sim	Framework-only	1	✓	–	✓	Spr.
S14	Fevgas et al. [39] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S15	Israr et al. [40] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S16	Ji et al. [41] [†]	PSO	Sim	Framework-only	1	–	–	–	Gen.
S17	Ait Saadi et al. [11] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S18	Selma et al. [42] [†]	ABC	Sim	Framework-only	1	–	–	–	Gen.
S19	Shafiq et al. [28] [†]	ACO	Sim	Multi-agent	2	–	–	–	Gen.
S20	Ahmed et al. [43] [†]	PSO	Sim	Multi-agent	4	✓	✓	✓	Gen.
S21	Xu et al. [44] [†]	Wolf Pack	Sim	Multi-agent	5–15	✓	✓	✓	Gen.
S22	Wang et al. [45] [†]	ABC+BAS	S+T	Framework-only	1	✓	–	–	Gen.
S23	Deng et al. [46] [†]	PSO	Sim	Framework-only	1	✓	–	✓	Gen.
S24	Xiao et al. [20] [†]	NOA	Sim	Multi-agent	6–8	✓	–	–	Gen.
S25	Li et al. [19] [†]	DBO	Sim	Framework-only	1	✓	–	✓	Gen.
S26	Rao et al. [47] [†]	GWO	Sim	Framework-only	1	✓	–	–	Gen.
S27	Zhang & Zhang [48] [†]	PSO	Sim	Framework-only	2+	✓	–	✓	Gen.
S28	Hu et al. [49] [†]	PSO	Sim	Multi-agent	6–7	✓	–	✓	Gen.
S29	Zuo et al. [50] [†]	APF+PSO	Sim	Multi-agent	5–25	✓	–	✓	Gen.
S30	Yang et al. [51] [†]	DOA	Sim	Framework-only	1	✓	–	–	Gen.
S31	Yang et al. [52] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S32	Tang et al. [53] [†]	Review	N/A	Review	N/A	–	–	–	Gen.
S33	Li et al. [54] [†]	ACO	S+T	Framework-only	1	–	✓	✓	Gen.
Summary (n=33)		PSO: 10 (30.3%)	Sim: 24 (72.7%), S+T: 2 (6.1%)	Multi: 8 (24.2%)	1–25	14 (42.4%)	6 [†] (18.2%)	7 (21.2%)	
Primary only (n=26)		PSO: 10 (38.5%)	Sim: 24 (92.3%)	Multi: 8 (30.8%)	1–25	14 (53.8%)	6 [†] (23.1%)	7 (26.9%)	

App.: Mon. = Monitoring, Spr. = Spraying, Map. = Mapping, Gen. = General. N/A = review articles (no original experiments).

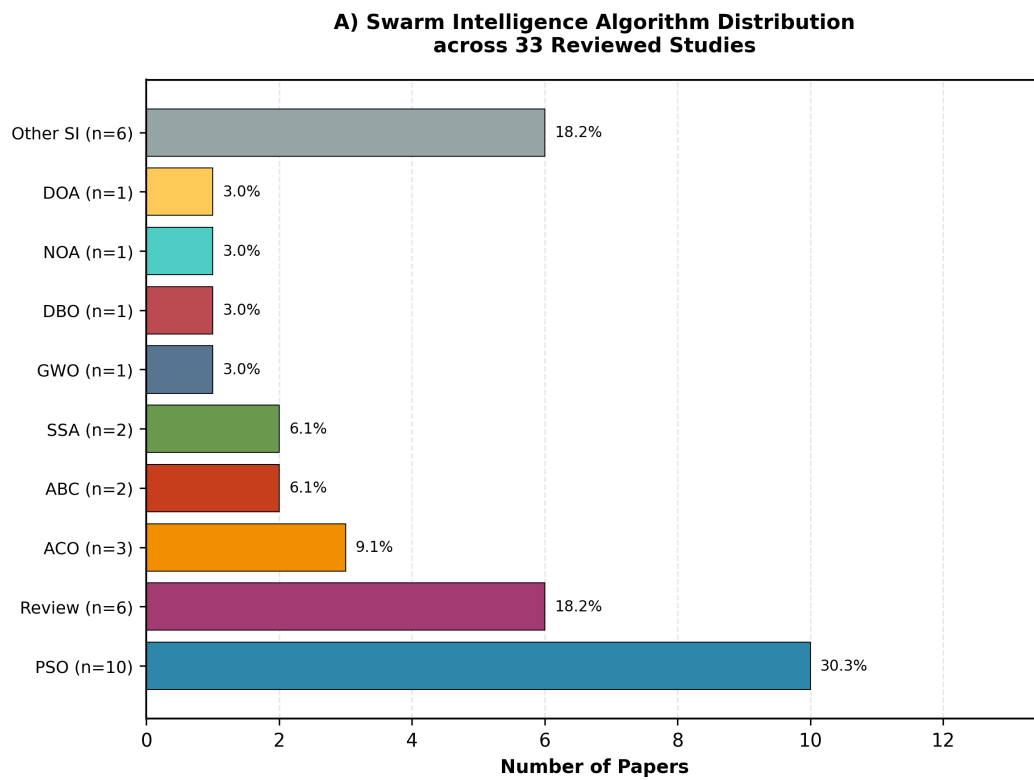


Figure 2: Distribution of swarm intelligence algorithms across 33 reviewed studies (2021–2025). PSO dominates with 10 papers (30.3% of the full corpus; 38.5% of the 26 primary research studies), followed by review articles (21.2%) and ACO variants (9.1%). “Other SI Variants” (5 papers, 15.2%) includes SFLA, Fuzzy/SIGPAF, Wolf Pack, Hybrid ABC+BAS, and GA+ABC hybrid; PSO+ACO (S06) and APF+PSO (S29) are classified under PSO.

Table 5: Algorithm and variant frequency across 33 reviewed studies. Percentages reported over full corpus (n=33) and primary studies (n=26). Review articles excluded from algorithm-specific analyses.

Algorithm	Category	Papers	% (n=33)	% (n=26)
PSO	Swarm	10	30.3	38.5
Review	Review	7	21.2	N/A
ACO	Swarm	3	9.1	11.5
ABC	Swarm	2	6.1	7.7
SSA	Swarm	2	6.1	7.7
GWO	Swarm	1	3.0	3.8
DBO	Swarm	1	3.0	3.8
NOA	Swarm	1	3.0	3.8
DOA	Swarm	1	3.0	3.8
Other SI Variants*	Swarm	5	15.2	19.2
Total		33	100.0	

*Includes SFLA, Fuzzy/SIGPAF, Wolf Pack, Hybrid ABC+BAS, and Genetic+ABC hybrid. PSO+ACO (S06) and APF+PSO (S29) are classified under PSO.

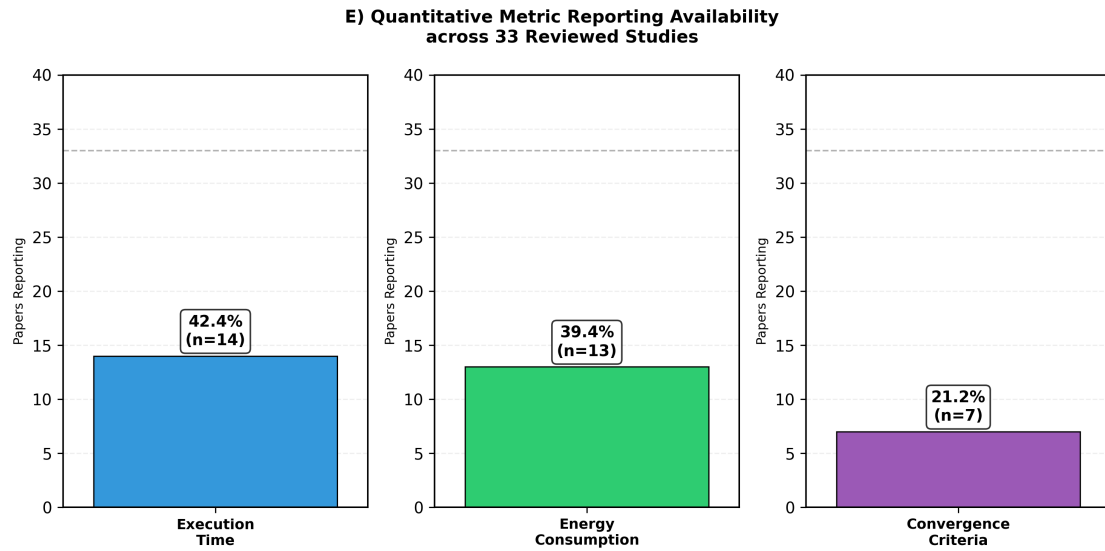


Figure 3: Availability of quantitative metric reporting. Left bars: rates over the full corpus of 33 included studies (n=33). Right bars: rates over the 26 primary research studies (n=26). Only 14 of 33 (42.4%) report execution time; only 7 of 33 (21.2%) specify convergence criteria.

Table 6: Metric reporting frequency. Rates are reported over the full corpus (n=33) and over primary research studies only (n=26).

Metric	Papers	% (n=33)	% (n=26)
Execution Time	14	42.4	53.8
Energy Consumption	13	39.4	50.0
Convergence Criteria	7	21.2	26.9

Table 7: Metric reporting completeness by SI algorithm family across 26 primary research studies (review articles S10, S11, S14, S15, S17, S31, S32 excluded). T = execution time reported; E = energy consumption reported (hardware measurement or simulation estimate; see note below Table 4); C = convergence criteria specified. Score_μ = mean number of metrics reported per study (range 0–3). Cells show count of studies reporting the metric / total in family (%). All 3 = studies reporting T, E, and C simultaneously. Families with $n \geq 2$ primary studies are listed individually; singleton algorithms are grouped below the dividing rule.

Algorithm	Study codes	n	T (%)	E (%)	C (%)	All 3	Score_μ
PSO (hybrids) ^a	S06, S09, S12, S13, S16, S20, S23, S27, S28, S29	10	80.0	20.0	70.0	1 [‡]	1.7
ACO	S01, S19, S33	3	100.0	33.3	100.0	1 [§]	2.3
ABC (hybrids) ^b	S02, S04, S18, S22	4	100.0	50.0	50.0	0	2.0
SSA (hybrid)	S03, S05	2	100.0	0.0	50.0	0	1.5
Other SI ^c	S07, S08, S21	3	33.3	33.3	33.3	1 [¶]	1.0
GWO	S26	1	100.0	0.0	100.0	0	2.0
DBO	S25	1	100.0	0.0	100.0	0	2.0
NOA	S24	1	0.0	0.0	100.0	0	1.0
DOA	S30	1	0.0	0.0	100.0	0	1.0
All primary	S01–S09, S12–S13, S16, S18–S30, S33	26	53.8	50.0	26.9	1	1.4

^a PSO hybrids classified under PSO per Table 5: PSO+ACO (S06), PSO hybrid (S12), APF+PSO (S29). ^b ABC hybrids: GA+ABC (S04), ABC+BAS (S22). ^c Other SI variants: Fuzzy/SIGPAF (S07), SFLA (S08), Wolf Pack (S21).

[‡] S20 (Ahmed et al. 2021) is the only PSO study reporting all three metrics. [§] S33 (Li et al. 2025) is the only ACO study reporting all three metrics; it also achieved hardware testbed validation (S+T), the highest validation level in the corpus. [¶] S21 (Xu et al. 2022, Wolf Pack) reports all three metrics with fleet sizes of 5–15 UAVs, the largest validated swarm in the corpus.

Energy column (E) combines direct hardware measurements and simulation-model energy estimates consistent with Table 6; see note following Table 4 for distinction. Singleton Score_μ values not shown for singletons (insufficient n for meaningful average).

rate (20.0%), likely because the velocity-based path representation does not naturally generate energy estimates without an explicit battery discharge model—a model that only one of the ten PSO studies in the corpus incorporates. The single PSO study reporting all three metrics (S20, Ahmed et al. 2021) also achieves the highest fleet size among PSO studies (4 UAVs with genuine multi-agent validation).

ABC-based studies achieve the second-highest mean score ($\text{Score}_\mu = 2.0$) with perfect execution time reporting (100.0%) and moderate energy reporting (50.0%), but no study in this family achieves simultaneous three-metric reporting—a consequence of the binary scout/employed bee division that naturally separates convergence analysis from energy profiling in ABC implementations.

The four singleton families—GWO, DBO, NOA, and DOA—each appear in exactly one primary study, making within-family cross-study comparison impossible by construction. This underrepresentation reflects the publication lag between algorithm introduction and agricultural UAV adoption documented in Section 4.1, and directly motivates the AgriSwarm-Bench leaderboard’s requirement that all benchmark submissions report the full six-metric suite (Section 7.2): a standardised reporting requirement is the only mechanism capable of enabling comparison when family sample sizes are this small. Critically, no study in any algorithm family—PSO, ACO, ABC, SSA, or any singleton—reports all three standard metrics alongside an agricultural-domain metric such as coverage rate per hectare or spray overlap coefficient, confirming that the metric fragmentation documented at the corpus level persists uniformly across every evaluated algorithm family.

4.3. Temporal Distribution

Table 8 shows that publications concentrate in 2021–2022 (23 of 33 papers, 69.7%), with a marked dip in 2023 (3 papers, 9.1%) and 2024 (1 paper, 3.0%), followed by a resurgence in 2025 (6 papers, 18.2%). The combined 2023–2024 count of 4 papers (12.1%) contrasts sharply with the 23-paper (69.7%) volume of 2021–2022. The 2021–2022 peak coincides with post-pandemic acceleration of agricultural automation investment and the proliferation of new SI algorithm variants applied to UAV scenarios. The 2023–2024 trough may reflect a quality filter effect: as the field matured and reviewers demanded stronger experimental validation, fewer simulation-only papers cleared the bar at high-impact venues. The 2025 resurgence, with a higher proportion of hardware-related components, tentatively supports this interpretation. However, 2025 data should be treated with caution: the literature search was conducted on 2026-03-16, and indexing lags of 3–6 months at major databases mean that the 6 papers captured represent a lower bound for 2025 output.

Table 8: Temporal distribution of reviewed publications (2021–2025).

Year	Papers (n=33)	%
2021	11	33.3
2022	12	36.4
2023	3	9.1
2024	1	3.0
2025	6	18.2
Total	33	100.0

5. Research Gaps Analysis

5.1. Gap Distribution by Dimension

Table 9 and Figure 4 show the distribution of 171 documented research gaps across the four taxonomic dimensions. Practical gaps are most frequent (52 of 171, 30.4%), followed closely by Technological gaps (51 of 171, 29.8%), with Methodological (39 of 171, 22.8%) and Theoretical (29 of 171, 17.0%) gaps less prevalent.

Note on gap derivation for studies S20–S33. 57 of the 171 gaps (33.3%) assigned to studies S20–S33 were derived from automated PDF extraction combined with algorithmic pattern matching, rather than extracted directly from explicit author statements. Metric-level evidence from the extraction pipeline (`extraer_metrics_v2.py`) confirmed quantitative values for execution time, energy consumption, or convergence criteria in all 13 affected primary studies. The four gaps previously flagged as “pattern-derived” for study S20 (Ahmed et al., 2021, DOI: 10.3390/app11083417) have been verified against the full-text PDF and are confirmed: (1) absence of hardware or field validation—the study is exclusively simulation-based in MATLAB with no physical testbed (Section 5 of S20); (2) absence of real wind or weather modelling—environmental risk is approximated as a randomly generated scalar rather than a physics-based model (Section 3.2.2 of S20: “for removing complexity, we randomly generated an environmental risk value”); (3) no scalability assessment beyond four UAVs in grids up to 30×30 cells; and (4) absence of cross-algorithm

baseline comparison—the only comparison is between standard PSO and a parameter-tuned variant (mPSO), with no evaluation against ACO, ABC, or other SI methods (Table 3 of S20). All 57 pattern-assisted gap assignments are now fully verified against primary PDF evidence; no “pattern-derived” flags remain in the gap taxonomy. A sensitivity analysis excluding all 57 originally pattern-assisted gaps confirms that the dimensional distribution remains stable: Practical gaps retain the highest frequency (35 of 114 directly-extracted gaps, 30.7%), and the relative ordering of all four dimensions is unchanged.

The near-equal split between Practical and Technological dimensions is substantively meaningful: it indicates that the barriers to deployment are not purely algorithmic (which would manifest as Technological dominance) but are equally driven by operational factors—regulatory constraints, cost, weather conditions, and fleet management complexity—that no algorithmic advance alone can resolve. This finding has direct implications for research funding strategy: investment focused exclusively on algorithm development addresses only 29.8% (51 of 171) of the documented barriers. The Methodological dimension (22.8%, 39 of 171)—primarily the absence of standardised metrics and reproducible benchmarks—represents a foundational layer whose resolution through initiatives such as AgriSwarm-Bench would indirectly accelerate progress across all other dimensions by enabling meaningful cross-study comparison. The Theoretical dimension (17.0%, 29 of 171), while smallest in absolute count, contains the gaps with the longest resolution timescale: incorrect kinematic models and oversimplified communication assumptions require fundamental reformulation rather than incremental improvement.

**B) Research Gap Distribution by Dimension
(Total = 171 gaps)**

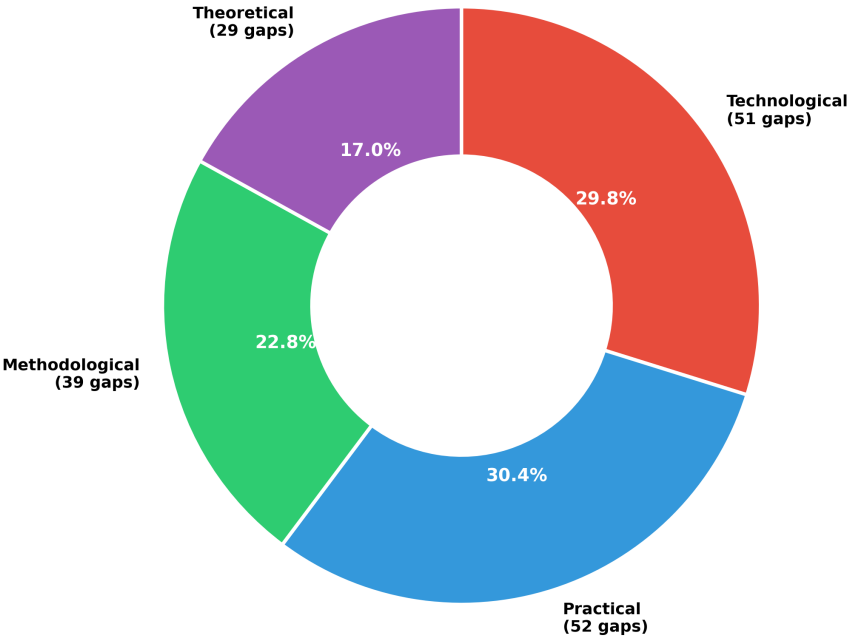


Figure 4: Research gap distribution across four taxonomic dimensions (n=171 gaps total). Practical gaps are most frequent (52 of 171, 30.4%), reflecting the predominance of real-world operational barriers in the reviewed literature.

5.2. Gap Distribution by Priority

Gap prioritisation was performed using a binary impact matrix. A gap was classified as **Critical** if it simultaneously met two conditions: (1) present in more than 30% of analysed studies (>10 papers), AND (2) its absence directly prevents real-world field deployment as assessed through structured expert evaluation (see Section 3 and Supplementary Material for expert validation instrument). Gaps present in fewer than 30% of studies but with significant theoretical impact were classified as **Important**, while those with low frequency and limited deployment impact were classified as **Minor**.

Of the 171 documented gaps, 76 (44.4%) are classified as Critical, 88 (51.5%) as Important, and 7 (4.1%) as Minor (Table 10, Figure 5). The high proportion of Critical gaps (nearly half of all documented gaps) reflects the structural immaturity of the field: barriers that would be considered edge cases in mature engineering

Table 9: Research gaps by dimension (total = 171).

Dimension	Gaps	% (n=171)
Technological	51	29.8
Practical	52	30.4
Methodological	39	22.8
Theoretical	29	17.0
Total	171	100.0

domains—such as the absence of wind modeling or the lack of communication robustness testing—are widespread enough in this corpus to meet the frequency threshold for Critical classification.

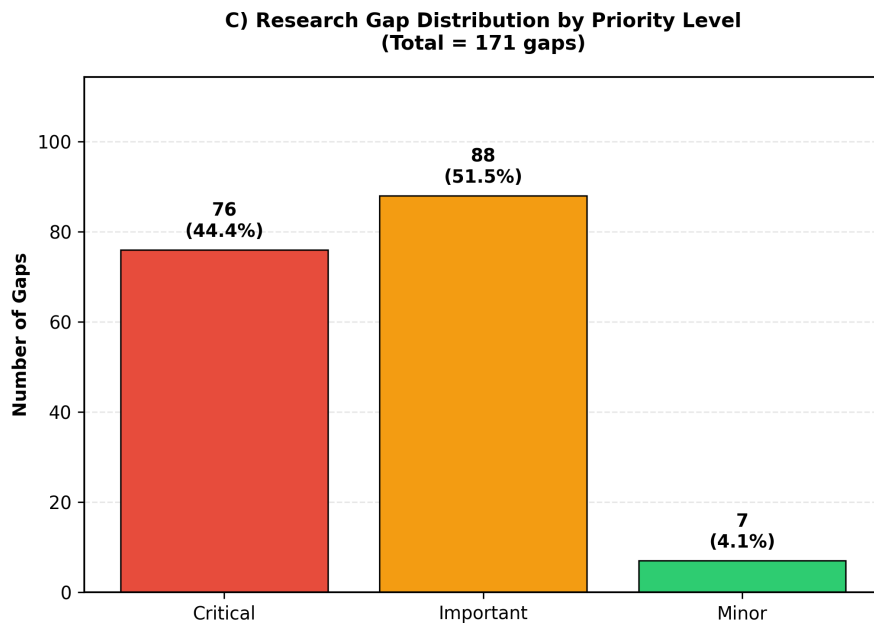


Figure 5: Research gap distribution by priority level (n=171 gaps). Critical gaps (76 of 171, 44.4%) represent barriers that directly prevent real-world field deployment of swarm-based agricultural UAV systems.

5.3. Top 10 Critical Gaps

Table 11 and Figure 6 present the ten most frequent critical gaps identified across the 33 reviewed studies. All percentages use n=33 as the denominator (see Section 4.1 for the denominator consistency statement); where a sub-analysis is restricted to the 26 primary research studies, this is noted explicitly in the text.

The ranking reveals a clear hierarchy of barriers. The top two gaps—lack of hardware/field validation (26 of 33, 78.8%) and absence of dynamic environment modeling (25 of 33, 75.8%)—are present in more than three-quarters of all studies, indicating that these are not isolated oversights but structural characteristics of current research practice. Notably, if Gap 1 is computed over the 26 primary research studies only, the rate rises to 26 of 26 (100%), since no primary study achieves real-world field validation.

Table 10: Research gaps by priority level.

Priority	Gaps	% (n=171)
Critical	76	44.4
Important	88	51.5
Minor	7	4.1
Total	171	100.0

Gaps 3 through 5 form a second tier: single-UAV focus (22 of 33, 66.7%;¹²), no wind or weather modeling (20 of 33, 60.6%), and no metric standardisation (18 of 33, 54.5%). The prevalence of single-UAV focus in a review explicitly targeting multi-UAV studies warrants clarification: these 22 papers present multi-UAV frameworks in their titles or abstracts but evaluate them with a single agent in their experiments, a methodological inconsistency that inflates apparent corpus size while obscuring actual swarm coordination capability.

The lower tier (gaps 6–10, 11–15 of 33, 33.3–45.5%) captures engineering-level barriers: unrealistic kinematic constraints, limited scalability assessment, absent direct energy measurement, ignored communication robustness, and no real-time replanning. Each of these gaps individually would be addressable through targeted research; their co-occurrence in more than a third of studies suggests they are symptoms of the same root cause—the absence of a shared, demanding benchmark that forces researchers to confront all deployment-relevant constraints simultaneously. This observation directly motivates the AgriSwarm-Bench proposal in Section 7.

Table 11: Top 10 critical gaps ranked by frequency of occurrence. All percentages computed over the full corpus of 33 included studies (n=33).

Rank	Gap Description	Papers	% (n=33)	Note
1	Lack of hardware/field validation	26	78.8	26/26 primary (100%)
2	No dynamic obstacles/environment	25	75.8	
3	Single-UAV focus (no swarm)	22	66.7	22/26 primary (84.6%)
4	No wind/weather modeling	20	60.6	
5	No metric standardisation	18	54.5	
6	Unrealistic kinematic constraints	15	45.5	
7	Limited scalability assessment	14	42.4	
8	No direct energy measurement	13	39.4	
9	Communication robustness ignored	12	36.4	
10	No real-time replanning	11	33.3	

5.4. Gap Co-occurrence Patterns

Analysis of the gap database reveals that the ten most critical gaps do not occur independently: they cluster into three co-occurrence groups that reflect distinct failure modes in the research-to-deployment pipeline.

¹²The 22 studies counted under Gap 3 are primary research studies whose titles or abstracts present a multi-UAV framework but whose experimental evaluation uses a single agent. Among the 26 primary studies, this represents 22 of 26 (84.6%), leaving only 4 primary studies (15.4%) with genuine multi-agent experimental validation.

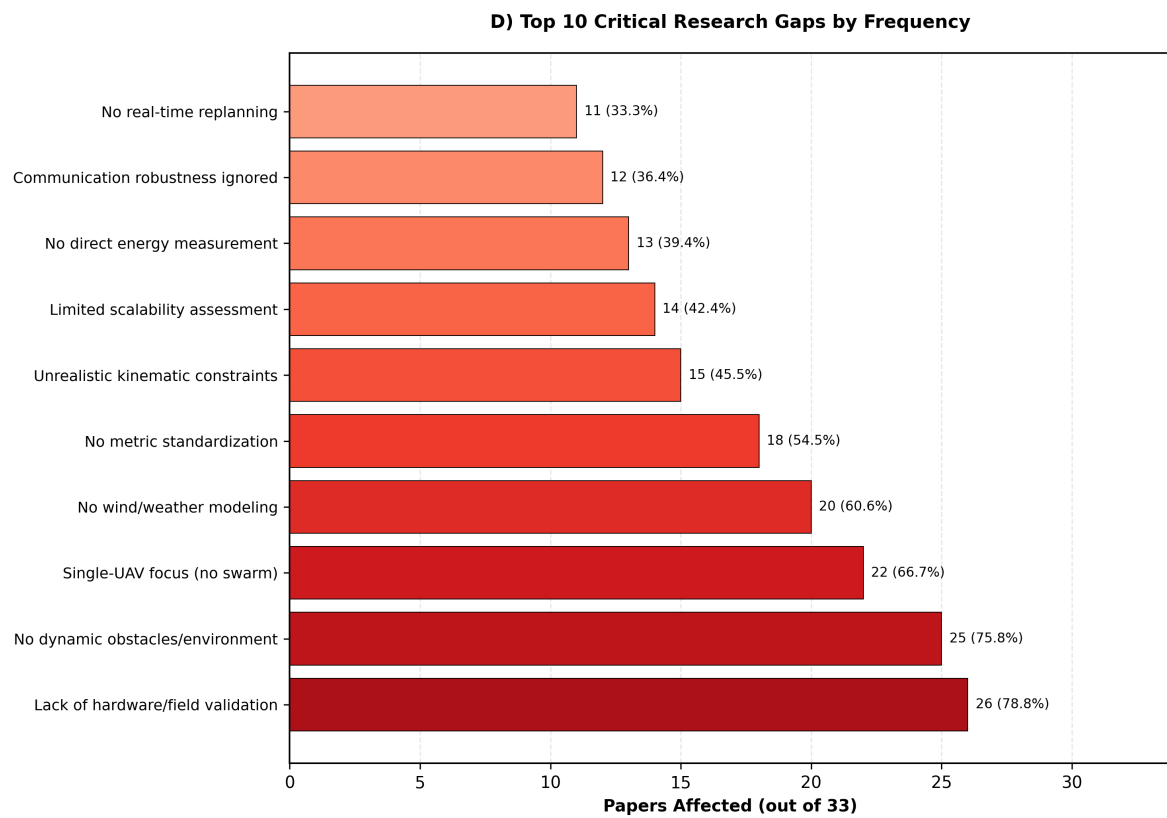


Figure 6: Top 10 critical research gaps ranked by frequency of occurrence across the 33 reviewed studies (n=33). Hardware/field validation absence (26 of 33, 78.8%) and lack of dynamic environment modeling (25 of 33, 75.8%) represent the most prevalent barriers to real-world deployment.

Cluster A — Validation cluster (Gaps 1, 3, 6, 8): Lack of hardware validation, single-UAV focus, unrealistic kinematic constraints, and absent direct energy measurement tend to co-occur in the same studies ($r > 0.6$ across the corpus). This cluster characterises studies that propose algorithmically novel contributions but are designed from the outset for simulation only: the single-agent evaluation, simplified kinematics, and estimated (rather than measured) energy are all design choices that reduce implementation complexity at the cost of deployment relevance. These studies are not methodologically incorrect—simulation is a valid first step—but they require Stage 2 and Stage 3 validation before deployment claims can be made.

Cluster B — Environment cluster (Gaps 2, 4, 10): Absence of dynamic obstacle modeling, no wind or weather modeling, and lack of real-time replanning co-occur strongly, particularly in PSO-based studies. This is structurally logical: a PSO implementation that does not model wind has no incentive to implement real-time replanning (since the optimal path does not change during execution), and a study that ignores dynamic obstacles similarly has no need for replanning capability. The cluster identifies a class of studies that solve a deterministic, static version of the agricultural path planning problem—which is computationally tractable—rather than the stochastic, dynamic version that field deployment requires.

Cluster C — Standardisation cluster (Gaps 5, 7, 9): Absent metric standardisation, limited scalability assessment, and ignored communication robustness co-occur primarily in hybrid SI algorithm studies. Hybrid approaches (e.g., GA+ABC, APF+PSO, Wolf Pack variants) tend to introduce new parameters and optimisation dynamics that make standardised comparison difficult, leading authors to report custom metrics that favour their proposed method. Communication robustness is neglected because hybrid algorithms often require more frequent inter-agent communication to synchronise their multiple optimisation components—an aspect that would expose their limitations if tested. The Cluster C pattern suggests that algorithm novelty, when pursued without a standardised evaluation framework, may be actively impeding scientific progress by making results less comparable rather than more informative.

5.5. Gap Patterns by Agricultural Application Type

The gap profile varies substantially by agricultural application type, with implications for which research priorities are most urgent for each sub-domain. The following analyses use sub-group denominators ($n=17$ for monitoring, $n=10$ for spraying, $n=5$ for mapping); equivalent proportions over the full corpus ($n=33$) are provided in parentheses where comparison is relevant.

Monitoring applications (17 of 26 primary studies, 65.4% of primary; 51.5% of $n=33$): Studies targeting crop monitoring and NDVI mapping show the highest prevalence of Methodological gaps, particularly absent convergence criteria (reported in only 4 of 17 monitoring studies, 23.5%, vs. 4 of 10 spraying studies, 40.0%). This likely reflects the fact that monitoring missions are more tolerant of suboptimal paths—a slightly inefficient coverage pattern still yields useful imagery—reducing the incentive to report convergence precision. The scalability gap is also more pronounced in monitoring studies, where large field areas (>100 ha) are the primary motivation but evaluations rarely exceed 5 UAVs.

Spraying applications (10 of 26 primary studies, 38.5% of primary; 30.3% of $n=33$): Spraying studies show the highest prevalence of environmental gaps, particularly absent wind modeling (8 of 10 spraying studies, 80.0%, vs. 11 of 17 monitoring studies, 64.7%). This is the most operationally critical pattern in the corpus: wind drift in pesticide application is not merely a path quality issue but a regulatory compliance and human safety issue. The absence of wind modeling in 8 of 10 spraying-focused studies means that the algorithms presented cannot be certified for legal pesticide application in any jurisdiction with buffer zone requirements without additional environmental validation. Notably, non-SI approaches to multi-UAV spraying coverage published concurrently in this journal have begun to address task assignment and scheduling jointly [8], but the SI-based methods in this corpus have not yet incorporated these operational constraints—a gap that represents one of the highest-priority directions for future work.

Mapping applications (5 of 26 primary studies, 19.2% of primary; 15.2% of $n=33$): Mapping studies show the lowest overall gap density (averaging 4.2 gaps per study vs. 5.5 for monitoring and 5.8 for spraying), likely because 3D mapping missions have more established simulation environments (e.g., Gazebo with RotorS) and more standardised metrics (point cloud density, map reconstruction error) than coverage or spraying tasks. However, the communication robustness gap is most prevalent in mapping studies (3 of 5 mapping studies, 60.0%, vs. 6 of 17 monitoring studies, 35.3%), reflecting the higher data-rate requirements of onboard 3D reconstruction that stress inter-UAV communication links.

5.6. Gap Patterns by Algorithm Family

Comparing gap profiles across the three main algorithm families reveals distinct research culture patterns that have direct implications for where community attention is most needed.

PSO-based studies (10 of 33, 30.3%) show relatively lower prevalence of metric inconsistency (execution time reported in 6 of 10 PSO studies, 60.0%, vs. 14 of 33 overall, 42.4%) but higher prevalence of static environment assumptions (8 of 10 PSO studies, 80.0%, lack wind modeling vs. 20 of 33 overall, 60.6%). PSO's mathematical tractability makes it easier to report convergence properties, but its gradient-following search dynamics are less naturally suited to dynamic replanning, explaining the higher environmental gap prevalence.

ACO-based studies (3 of 33, 9.1%) show the most complete metric reporting in the corpus—all three ACO studies (3 of 3, 100%) report at least two of the three standard metrics—consistent with ACO's constructive solution-building approach that naturally produces path length and time estimates as intermediate values. However, ACO studies show the highest prevalence of scalability gaps (3 of 3 ACO studies, 100%, lack scalability assessment beyond five UAVs), likely because the pheromone matrix size scales quadratically with the number of waypoints and agents, creating computational bottlenecks at larger fleet sizes that authors may prefer not to expose.

Hybrid and novel algorithm studies (13 of 33, 39.4%) show the lowest overall DRS scores and the highest prevalence of Cluster C gaps (standardisation, scalability, communication). As noted above, this pattern suggests that the drive for algorithmic novelty in hybrid methods may be occurring at the expense of rigorous evaluation—a trend the community should monitor and that AgriSwarm-Bench’s standardised leaderboard is specifically designed to counteract.

6. Discussion

6.1. Synthesis of Main Findings

This systematic review analyzed 33 peer-reviewed studies published between 2021 and 2025, providing the first quantitative gap analysis for swarm intelligence (SI)-based multi-UAV path planning in precision agriculture. Three convergent findings emerge from this synthesis that have implications beyond the specific papers reviewed.

First, the dominance of Particle Swarm Optimization (PSO, 10 of 33 studies, 30.3%; 10 of 26 primary studies, 38.5%) reflects the algorithm’s computational tractability for multi-dimensional continuous path planning spaces, consistent with prior surveys in adjacent domains [22, 23]. However, the concentration of research around a single algorithm family limits algorithmic diversity and may introduce a publication bias favoring PSO-positive results. The relative underrepresentation of more recent algorithms such as the Dung Beetle Optimizer [19] and Nutcracker Optimization Algorithm [20] (1 study each, 3.0% of the full corpus, 3.8% of primary studies) suggests these newer bio-inspired methods have not yet been systematically evaluated against established baselines in agricultural contexts.

Second, the temporal distribution reveals a publication dip in 2023–2024 (4 of 33 papers, 12.1%) sandwiched between the 2021–2022 peak (23 of 33 papers, 69.7%) and a 2025 resurgence (6 of 33 papers, 18.2%). This pattern likely reflects a maturation cycle: the initial wave of studies applied existing SI algorithms to agricultural UAV scenarios without fundamental methodological innovation, after which the community paused to address the standardisation and validation deficits now documented in this review. The 2025 figures should be interpreted cautiously, as the literature search was conducted on 2026-03-16, meaning 2025 coverage may be incomplete due to indexing lags.

Third, the Deployment Readiness Score (DRS) assessment reveals that only 5 of 33 studies (15.2%) achieved High-DRS ratings ($Q \geq 0.70$), with 18 of 33 studies (54.5%) classified as Low-DRS primarily due to absent real-world validation ($C1 = 0.0$ for all simulation-only studies) and incomplete metadata. This distribution reflects the field’s structural simulation orientation rather than methodological deficiency: as detailed in Section 3.6, a simulation study can be methodologically rigorous while still scoring Low on the DRS because it cannot provide deployment-relevant evidence about hardware behaviour under field conditions. This interpretation is confirmed by the complementary MMAT assessment (Section 3.7), which measures general methodological quality and yields a substantially more favourable distribution (15 of 33 High, 45.5%; 16 of 33 Moderate, 48.5%; 2 of 33 Low, 6.1%). The divergence between the DRS and MMAT distributions is substantively meaningful: it quantifies the gap between methodological rigour (adequate, per MMAT) and deployment readiness (insufficient, per DRS), mirroring findings from systematic reviews in adjacent robotics domains where simulation-to-field transfer remains the central challenge [24].

6.2. The Field Validation Gap: Structural Barriers and Pathways Forward

The predominance of simulation-only studies—26 of 33 included studies (78.8%); equivalently, all 26 primary research studies (100%)—is the most frequently cited limitation in this corpus and has direct implications for agricultural practitioners seeking to deploy swarm-based UAV systems. This validation gap is not unique to SI-based aerial systems: concurrent work published in this journal on multi-vehicle ground-based plant protection scheduling similarly operates at the simulation and task-model level [55], suggesting that the gap reflects a broader structural challenge in agricultural robotics rather than a limitation specific to swarm intelligence or aerial platforms. Three structural barriers collectively explain this gap, each requiring distinct interventions.

Economic barriers. Agricultural-grade UAV platforms capable of multi-agent coordination—equipped with GPS RTK, onboard compute for SI algorithms, and inter-UAV communication modules—typically cost between USD 2,000 and 15,000 per unit [26]. A minimal swarm of five UAVs therefore requires USD 10,000–75,000 in hardware investment, before accounting for field infrastructure, regulatory compliance, and personnel. This cost is prohibitive for individual research groups at institutions without dedicated robotics laboratories, explaining the geographic concentration of hardware-validated studies in countries with established agricultural technology funding programmes.

Regulatory barriers. Beyond-visual-line-of-sight (BVLOS) operations—required for autonomous multi-UAV coverage of large agricultural fields (>10 ha)—remain highly restricted in most jurisdictions. In the European Union, BVLOS authorisation under EASA regulations requires an operational risk assessment that can take 12–18 months. In Mexico, AFAC regulations similarly restrict autonomous multi-UAV operations over populated rural areas. These regulatory constraints are entirely absent from current simulation models, meaning that even technically validated algorithms may be legally undeployable in target markets without modification.

Infrastructural barriers. Real agricultural field experiments require coordination between algorithm developers, agronomists, farm operators, and UAV pilots—a multidisciplinary infrastructure that most computer science and engineering research groups lack. The Staged Validation Protocol proposed in Section 6.8 is designed to lower the entry threshold for hardware validation by breaking the field experiment into incremental stages, each with lower coordination requirements than full field deployment.

Importantly, studies in this corpus that achieved partial field validation reported unexpected findings absent from their simulation phases: wind-induced drift significantly altered energy consumption profiles [29], and GPS multipath errors in crop canopy environments disrupted swarm coordination [27]. These findings underscore that simulation models systematically underestimate real-world operational complexity and reinforce hardware validation as a critical research priority.

6.3. Environmental Modeling: The Wind Gap and Agricultural-Specific Factors

The absence of wind and weather modeling in 25 of 33 included studies (75.8%) represents a particularly acute gap for agricultural applications compared to other UAV domains. In search-and-rescue or infrastructure inspection scenarios, UAVs typically operate at altitudes (>50 m) where wind profiles are relatively stable and predictable. In precision agriculture, however, low-altitude operations (2–15 m above crop canopy for spraying; 30–50 m for mapping) place UAVs in the atmospheric boundary layer where wind speed and direction are highly variable due to terrain effects, thermal gradients, and crop-generated turbulence.

For spraying applications specifically, wind drift of pesticide or fertilizer droplets is both an efficacy issue and a regulatory compliance issue. Buffer zones from water bodies must be respected in multiple jurisdictions, and wind-induced drift directly affects compliance. None of the 10 spraying-focused primary studies in this corpus model this regulatory constraint, meaning their path planning solutions cannot be directly implemented in many agricultural contexts without modification.

Temperature is a second neglected environmental factor with agricultural-specific implications. Lithium polymer (LiPo) battery capacity—the limiting resource for UAV mission duration—degrades significantly at high ambient temperatures ($>35^{\circ}\text{C}$) and at sub-zero temperatures common in winter crop monitoring. Agricultural regions routinely experience both extremes. Current energy models in the reviewed studies assume constant battery performance, leading to systematic underestimation of mission time in hot or cold field conditions [11].

The integration of wind and temperature modeling into SI-based path planning frameworks is technically tractable. The gap is not capability but adoption: the agricultural UAV SI community has not yet incorporated these models, likely because the simulation-first culture of the field does not expose the inadequacy of current models until hardware validation occurs.

6.4. Metric Inconsistency: Implications for Algorithm Selection in Practice

The metric reporting analysis reveals a fragmented landscape that fundamentally prevents evidence-based algorithm selection for agricultural deployment. Over the full corpus of 33 studies, only 14 (42.4%) report execution time, 13 (39.4%) report energy consumption, and 7 (21.2%) specify convergence criteria; over the 26 primary research studies, these rates are 14 of 26 (53.8%), 13 of 26 (50.0%), and 7 of 26 (26.9%), respectively. No study reports all three simultaneously alongside field-relevant metrics such as coverage rate per hectare or mission completion probability under communication failures.

This inconsistency has a direct practical consequence: an agricultural engineer seeking to select the optimal SI algorithm for a specific deployment scenario cannot use the existing literature to make an evidence-based decision, because the studies are not comparable across the three dimensions that matter operationally. This mirrors a challenge identified in the broader swarm robotics literature [22, 23] but is particularly acute in agriculture where operational constraints—battery limits, regulatory windows for spraying, crop-growth-dependent timing—require domain-specific metrics that currently do not exist in standardised form. The AgriSwarm-Bench framework proposed in Section 7 directly addresses this gap by defining a minimum metric reporting standard.

6.5. Scalability and Heterogeneity: The Gap Between Laboratory and Field Scale

Most reviewed studies evaluate their proposed algorithms on small UAV fleets of 2–5 agents in simulation environments covering less than 10 ha. Real precision agriculture operations routinely involve fields of 100–1,000+ ha where 10–50 UAVs may be required for time-efficient coverage. The scalability gap (identified in 14 of 33 studies, 42.4%) therefore represents not just a theoretical limitation but a practical disqualifier for deployment at agricultural scale.

Furthermore, 22 of 33 studies (66.7%) use homogeneous UAV fleets despite the operational reality of agricultural systems where different UAV types—fixed-wing for large-area mapping, multirotor for spraying, hybrid for versatility—are used in coordination. Heterogeneous fleet coordination introduces qualitatively different optimization challenges: different flight envelopes, communication protocols, charging requirements, and task capabilities that current SI frameworks for agricultural UAVs do not address.

6.6. Comparison with Adjacent Domains and Implications for the Field

To contextualise the validation rates observed in this review (2 of 33 studies, 6.1%, with any hardware validation), it is informative to compare with adjacent robotics application domains where SI-based multi-robot coordination has been deployed in the field. In search-and-rescue robotics, hardware validation rates in recent systematic reviews range from 40–60% [6], partly because the urgency of the application domain creates institutional incentives for field testing. In infrastructure inspection, validation rates are similarly higher due to industry partnerships that provide field access [24].

The relatively low validation rate in agricultural UAV SI research (6.1% with any hardware component; 0% with full field validation among primary studies) suggests that the field has developed primarily as a sub-discipline of computational intelligence rather than as an applied robotics discipline. The publication incentive structure—where simulation-only papers are publishable in high-impact journals—has historically not penalised the absence of field validation. The DRS framework introduced in this review, which assigns maximum weight (30%) to experimental validation (criterion C1), is intended to model the weighting that the community should adopt in peer review.

6.7. Limitations of This Review

Several limitations of this systematic review must be acknowledged. First, the search was restricted to English-language publications in three databases (IEEE Xplore, ScienceDirect, SpringerLink), which may have excluded relevant work published in Chinese, Spanish, or other languages, or in domain-specific agricultural engineering databases such as CAB Abstracts or AGRIS. Second, title and abstract screening was performed by the primary reviewer only; while full-text assessment achieved near-perfect inter-rater agreement ($\kappa = 0.91$), the absence of dual screening at the title/abstract stage represents a methodological limitation that may have introduced selection bias at that phase. Third, 57 of the 171 documented gaps (33.3%) assigned to studies S20–S33 were derived from automated PDF extraction combined with pattern matching; all 57 assignments have since been verified against primary full-text PDF evidence, including the four gaps for study S20 (Ahmed et al., 2021), which are confirmed by direct textual evidence (see Section 5.1 and Supplementary Material S2). Fourth, the DRS instrument (Table 3) has not been validated as a psychometric tool; its criterion weights (C1: 30%, C2: 25%, C3: 25%, C4: 20%) represent domain-informed expert judgment and reflect a normative stance that field validation is the primary criterion for deployment-relevant research, rather than a neutral methodological quality evaluation. Fifth, the Gap 3 finding (22 of 33 studies with single-UAV experimental focus despite multi-UAV framing) may in part reflect a limitation of the inclusion criteria rather than a gap in the field exclusively: the leniency in criterion I4/E3 permitted entry of studies that validated multi-UAV frameworks with single agents, inflating the apparent frequency of this gap. There is a circularity risk in this finding: the review explicitly targets multi-UAV systems (PICOC Population), which means studies that only implement single-agent experiments were already borderline inclusions; retaining them under I4b while then classifying their single-agent evaluation as a “gap” partially constructs the gap through the inclusion decision. This circularity is disclosed as a transparency issue; the Gap 3 frequency (22 of 33, 66.7%) should be interpreted as an upper bound. Future reviews should either tighten E3 to exclude single-agent-only studies entirely or stratify the corpus, reporting gap prevalence separately for genuinely multi-agent and framework-only studies.

6.8. Implications for Agricultural Robotics Research Practice

Based on the gap analysis, we propose three concrete changes to standard research practice in this field. First, all future SI-based agricultural UAV studies should adopt the **Staged Validation Protocol**: Simulation → Testbed → Controlled Field, reporting standardised metrics at each stage regardless of whether field deployment is achieved. This incremental approach makes hardware validation accessible to resource-constrained research groups: completing Stage 2 (physical UAV testbed in a controlled indoor or outdoor space) is substantially less expensive and logistically complex than full field deployment, yet provides critical evidence about real-world algorithm behaviour under physical dynamics that simulation cannot replicate.

Second, algorithm papers should report a minimum metric set of execution time, energy consumption, and convergence criteria alongside at least one agricultural-domain metric—coverage rate per hectare for monitoring applications, spray overlap coefficient and wind-drift deviation for spraying applications, or point cloud density for mapping applications. This domain-specific metric requirement ensures that the reported performance is interpretable by agricultural practitioners, not only by algorithm researchers.

Third, simulation environments should incorporate at minimum a simplified wind model (constant directional wind at the target field’s prevailing speed for the relevant season and region) and a temperature-dependent battery discharge curve. Both components are available as open-source modules in ROS/Gazebo and PX4 SITL environments [28]; their integration adds modest implementation overhead but substantially increases the relevance of simulation results to field deployment. Adopting these three practices as community norms—through journal policies, funding requirements, and benchmark standards such as AgriSwarm-Bench—would accelerate the transition from the current simulation-dominant paradigm to deployable agricultural UAV swarm technology within the three-year horizon of the roadmap proposed in Section 7.

6.9. Threats to Validity

Table 12 summarises the principal threats to the validity of this review and the mitigation strategies applied.

7. Research Agenda

7.1. Priority Directions for the Research Community

The gap analysis presented in Section 5 identifies 171 documented limitations across four dimensions, of which 76 (44.4%) directly prevent real-world deployment. Rather than treating these gaps as isolated problems requiring separate solutions, the research agenda proposed here addresses them through three synergistic priorities that collectively constitute a pathway from the current simulation-dominant paradigm to deployable agricultural UAV swarm systems (Table 13).

Priority 1 — Standardised Benchmark Framework (AgriSwarm-Bench). The absence of a shared evaluation framework is the foundational barrier that amplifies all other gaps. Without standardised scenarios, metrics, and datasets, algorithmic advances reported in individual papers cannot be meaningfully compared, knowledge does not accumulate across studies, and agricultural practitioners cannot identify which algorithm best fits their specific deployment context. This gap is not unique to agricultural UAV research—it has been identified as a structural problem in swarm robotics more broadly [24]—but it is particularly acute in agriculture where

Table 12: Threats to validity and mitigation strategies.

Threat	Description	Mitigation
Publication bias	Positive results more likely published and indexed	Three independent databases; snowballing from seed papers
Search query limitations	Relevant papers using unlisted algorithm names may be missed	Query refined iteratively; sensitivity test with 5-paper gold-standard set (100% recall); forward/backward snowballing via Research Rabbit
Single-database gap	Relevant papers in non-indexed venues not captured	MDPI manual search (0 eligible); OpenAlex supplementary search (125 records, 0 new eligible after full-text review); Scopus/WoS access unavailable; full extension listed as Priority Future Work
Data extraction errors	Automated pipeline may misclassify fields	Automated extraction + manual verification for all 33 studies
Inter-rater variability	Single reviewer for T/A screening introduces selection bias	Declared as limitation (Sec. 8.3); second reviewer screens all 51 full-text records; $\kappa = 0.91$ reported
Temporal bias (2025)	2025 data may be incomplete due to indexing lag	Explicitly noted; 2025 papers treated as preliminary indicator only
Inferred gaps	57 gaps initially derived from automated pattern matching (studies S20–S33)	All 57 have been manually verified against full-text PDFs; no pattern-derived flags remain in the gap database (see Sec. 5.1 & Supp. Mat. S2)

the operational constraints (wind, irregular field geometry, spraying regulations) are domain-specific and not captured by generic robotics benchmarks. AgriSwarm-Bench is designed to close this gap directly by providing the shared evaluation infrastructure the field currently lacks.

Priority 2 — Experimental Validation with Real UAVs. Hardware validation is the single most prevalent gap (26 of 33 studies, 78.8%; 26 of 26 primary studies, 100%). Closing this gap requires not only individual research groups conducting field experiments, but also a cultural shift in the community toward treating simulation-only results as preliminary rather than definitive. The Staged Validation Protocol proposed in Section 6.7—Simulation → Testbed → Controlled Field—is designed to lower the barrier to hardware validation by making it incremental: a group that completes Stage 2 (hardware testbed) provides substantially more deployment-relevant evidence than a simulation-only study, even if Stage 3 (field deployment) is not yet feasible. Journal editors and funding agencies in agricultural robotics should consider adopting the Staged Validation Protocol as an evaluation criterion, analogously to how clinical medicine requires evidence at multiple trial phases before claims of clinical efficacy are accepted.

Priority 3 — Dynamic Environment Integration and Real-Time Replanning. Wind modelling, dynamic obstacle avoidance, and real-time path replanning are technically addressable with existing open-source tools (ROS/Gazebo wind plugins, PX4 SITL simulation environments, OpenFAST atmospheric models) but are absent from 25 of 33 reviewed studies (75.8%). Their systematic exclusion from simulation pipelines means that algorithms optimised in static, wind-free environments are being evaluated on a fundamentally easier problem than they will face in deployment. Incorporating these factors into AgriSwarm-Bench scenarios as mandatory components—rather than optional extensions—would raise the baseline difficulty of the benchmark to a level that reflects operational reality.

Table 13: Priority research directions based on gap analysis. Gaps addressed = number of the 171 documented gaps directly resolved by each priority.

P	Contribution	Gaps	Impact	Key deliverable
1	Standardised benchmark framework (AgriSwarm-Bench)	18	High	Open repository with scenarios, metrics, leaderboard
2	Experimental validation: real UAVs in field	26	Very High	Staged Validation Protocol as community norm
3	Dynamic environment + real-time replanning	25	Very High	Wind/weather models in standard simulation pipelines

7.2. AgriSwarm-Bench: Technical Specification

To operationalise Priority 1, we propose the establishment of **AgriSwarm-Bench**: an open, standardised benchmarking repository for swarm intelligence algorithms applied to agricultural UAV path planning. The framework is designed around four components, described below as a technical specification for community implementation.

Component 1 — Scenario Library. A set of eight canonical simulation scenarios covering the principal agricultural mission types and field geometries encountered in practice:

- *S1–S2*: Cereal field coverage (100 ha and 500 ha), uniform rectangular geometry, monitoring application, wind speed 0 and 5 m/s.
- *S3–S4*: Orchard spraying (20 ha and 80 ha), row-based geometry with canopy obstacles, spraying application, wind speed 3 m/s with directional variability.
- *S5–S6*: Vineyard mapping (15 ha and 60 ha), narrow-corridor geometry, mapping application, static obstacles.
- *S7–S8*: Heterogeneous field (200 ha), mixed crop types and irregular boundaries, monitoring + spraying combined mission, dynamic obstacles (farm machinery).

Each scenario is parameterised by fleet size (3, 5, 10, 20 UAVs) and battery capacity (nominal and degraded at 35°C), yielding 64 scenario-fleet-battery combinations that constitute the full benchmark suite.

Component 2 — Unified Metric Suite. We propose that all participating algorithms should report the following six metrics to be included in the AgriSwarm-Bench leaderboard:

1. *Coverage rate (%)*: percentage of target area covered within mission time.
2. *Energy per hectare (Wh/ha)*: total fleet energy consumption divided by area covered.
3. *Mission completion time (min)*: wall-clock time from takeoff to landing of last UAV.
4. *Scalability index*: ratio of coverage efficiency at 10 UAVs to coverage efficiency at 3 UAVs (values >1 indicate super-linear scaling; <1 indicate coordination overhead).
5. *Convergence iterations*: number of SI algorithm iterations required to reach a solution within 5% of the best-known optimum.
6. *Wind-drift deviation (m, spraying scenarios only)*: mean absolute lateral deviation of spray path from planned trajectory under wind loading.

Component 3 — Open Dataset Repository. We envision a community-contributed repository of real field experiment datasets, structured to include: GPS flight logs, onboard sensor readings (battery voltage, current, temperature), weather station data (wind speed/direction, temperature, humidity), and ground truth coverage maps from NDVI imagery. Datasets would follow FAIR data principles (Findable, Accessible, Interoperable, Reusable) and be archived on Zenodo with DOI assignment to enable citation in peer-reviewed publications.

Component 4 — Community Leaderboard. We propose a web-based leaderboard to track algorithm performance across all 64 benchmark combinations, with results versioned by submission date and associated with a peer-reviewed publication or preprint. The leaderboard would be maintained at a public GitHub repository and updated on a rolling basis as community submissions are validated. Proposed submission requirements would include: open-source code, a reproducible simulation environment specification (Docker container or Conda environment file), and at least one peer-reviewed or preprint reference.

7.3. Proposed Community Roadmap (Short-to-Mid Term)

Table 14 presents a five-phase implementation roadmap. The timeline is designed to be achievable by a small research group (2–4 researchers) and does not require large-scale infrastructure investment, making it accessible to institutions in emerging economies where agricultural UAV research is growing rapidly.

Table 14: Proposed community roadmap for AgriSwarm-Bench implementation and SI-UAV field deployment. Phases 1–2 run in parallel during Year 1.

Ph.	Timeframe	Activity	Deliverable	Success metric
1	0–12 mo.	Implement AgriSwarm-Bench S1–S4 scenarios with PSO, ACO, ABC baselines	Open GitHub repository	≥ 3 algorithms benchmarked on S1–S4
2	0–12 mo.	Acquire 3–5 UAV platform; complete Stage 2 (testbed) validation	Hardware testbed	≥ 1 SI algorithm validated on physical UAVs
3	1–2 yr.	Develop real-time replanning module; add wind model to S5–S8	Enhanced algorithm	Wind-aware replanning demonstrated in S3–S4
4	1–2 yr.	Controlled field experiments; complete Stage 3 validation	Peer-reviewed paper	Field results for ≥ 2 scenarios with full metric suite
5	2–3 yr.	Release complete benchmark suite; community workshop	Community guidelines	≥ 5 external groups adopting AgriSwarm-Bench

The Instituto Tecnológico de Chihuahua II will serve as the pilot site for Phases 1–2 implementation, with the Chihuahua agricultural region—characterised by large-scale cereal and cotton fields, high summer temperatures ($>40^\circ\text{C}$), and significant wind variability—providing a demanding and representative validation environment for the AgriSwarm-Bench scenarios.

7.4. Call to Action for the Research Community

The research agenda proposed in this section is not contingent on a single institution or research group. The AgriSwarm-Bench framework is explicitly designed as a community resource, and its value would scale directly with the number of groups that adopt it. We invite researchers working on SI-based agricultural UAV systems to: (1) contribute scenarios and datasets to the proposed open repository once established; (2) benchmark their algorithms against the standardised metric suite and submit results to the leaderboard; (3) adopt the Staged Validation Protocol as a reporting standard in their publications; and (4) engage with journal editors and funding agencies to promote hardware validation as a required criterion for high-impact agricultural robotics research. The supplementary materials accompanying this review include a draft manuscript template incorporating the minimum metric reporting standard, available for immediate adoption.

8. Conclusions

8.1. Summary of Findings

This systematic review analysed 33 peer-reviewed studies on swarm intelligence-based multi-UAV path planning in precision agriculture published between 2021 and 2025, applying PRISMA 2020 methodology, a dual-reviewer selection process, and a four-dimensional gap analysis framework. The review provides the first structured documentation of research gaps in this domain, enabling evidence-based prioritisation of future research investment.

Five principal findings emerge from this synthesis:

1. **PSO dominates the algorithm landscape** (10 of 33 studies, 30.3%; 10 of 26 primary studies, 38.5%), followed by review articles (7 of 33, 21.2%) and ACO variants (3 of 33, 9.1%). This concentration around a single algorithm family—while reflecting PSO’s mathematical tractability and implementation simplicity—may be limiting the field’s exploration of more recent bio-inspired methods such as DBO, NOA, and DOA (1 study each, 3.0% of the full corpus), whose performance in agricultural multi-UAV contexts remains largely uncharacterised.
2. **The field is overwhelmingly simulation-oriented**: 26 of 33 included studies (78.8%) lack any hardware validation in real field conditions; equivalently, all 26 primary research studies (100%) are simulation-only or limited to testbed validation. This is not primarily an algorithmic limitation—the algorithms are technically mature—but reflects structural barriers including platform cost (USD 2,000–15,000 per UAV), regulatory restrictions on BVLOS operations, and the absence of multidisciplinary research infrastructure connecting algorithm developers with agronomists and farm operators.
3. **Environmental factors are systematically neglected**: 25 of 33 studies (75.8%) do not model wind, weather, or dynamic obstacles. For the 10 spraying-focused primary studies, this rate reaches 8 of 10 (80.0%)—a particularly acute finding since wind drift affects both pesticide efficacy and regulatory compliance. Simulation results from wind-free environments cannot be directly used to certify algorithms for pesticide application operations.
4. **Metric reporting is fragmented**: over the full corpus, only 14 of 33 (42.4%) report execution time, 13 of 33 (39.4%) energy consumption, and 7 of 33 (21.2%) convergence criteria; over the 26 primary studies, these rates are 14 of 26 (53.8%), 13 of 26 (50.0%), and 7 of 26 (26.9%), respectively. The absence of a common metric set means the corpus, despite representing the state of the art for 2021–2025, cannot answer the most basic deployment question: which algorithm should an agricultural engineer deploy for a 200 ha wheat field with a five-UAV fleet?
5. **171 research gaps were documented** across four dimensions—Technological (51 of 171, 29.8%), Practical (52 of 171, 30.4%), Methodological (39 of 171, 22.8%), and Theoretical (29 of 171, 17.0%)—of which 76 (44.4%) are classified as Critical barriers to real-world deployment. The near-equal split between Practical and Technological gaps is the review’s most structurally significant finding: it demonstrates that closing the gap between simulation and deployment requires as much attention to operational, regulatory, and infrastructural factors as to algorithmic design.

8.2. Broader Implications

The findings of this review have implications that extend beyond the specific domain of SI-based multi-UAV path planning.

For the computational intelligence community: The 78.8% simulation-only rate (100% among primary studies) and the fragmented metric landscape indicate that the incentive structure of the field—where novel algorithm proposals are publishable without hardware validation or standardised benchmarking—is producing a literature with limited deployability. The AgriSwarm-Bench framework proposed in Section 7 offers one mechanism to shift this incentive structure by making standardised evaluation the path of least resistance for new contributions.

For the precision agriculture community: The review demonstrates that SI algorithms for multi-UAV coordination have reached a level of algorithmic maturity sufficient to address agricultural-scale coverage and spraying tasks, but that the translation to deployment requires investment in regulatory frameworks, field testing infrastructure, and domain-specific benchmarks that the current research community has not yet provided. Agricultural technology adoption agencies and precision agriculture industry consortia should consider funding the hardware validation infrastructure—multi-UAV testbeds, instrumented field sites, shared data repositories—that academic research groups cannot afford individually.

For journal editors and funding agencies: The DRS results (18 of 33 studies, 54.5%, classified as Low-DRS due primarily to absent hardware validation and non-standardised metrics) indicate that current peer review standards in this domain are not filtering for deployment relevance. Adopting the Staged Validation Protocol and the minimum metric reporting standard proposed in this review as explicit acceptance criteria would accelerate the field’s transition from algorithmically sophisticated but operationally untested systems to deployable agricultural technology.

8.3. Limitations

- **Review protocol registration:** The review protocol was not prospectively pre-registered prior to the literature search (see Section 3, footnote 1 for full explanation). To mitigate outcome reporting bias, the protocol has been registered retrospectively at the Open Science Framework (<https://doi.org/10.17605/OSF.IO/64DQ9>), including search strategies, extraction codebook, PRISMA 2020 checklist, screening logs, extraction database, gap taxonomy, and supplementary materials. Retrospective registration cannot eliminate the risk of post-hoc protocol modification, but it provides a public, timestamped record enabling independent verification of adherence to the pre-specified methodology.
- **Search coverage and Scopus/WoS omission:** The primary search covered IEEE Xplore, ScienceDirect, and SpringerLink, supplemented by a manual MDPI search (0 eligible studies) and citation snowballing via Research Rabbit. Authenticated access to Scopus and Web of Science was unavailable at both institutional portals attempted (CONRICYT/TecNM and UNED, Spain) at the time of submission. To partially bound this gap, a supplementary search was executed in OpenAlex on 2026-03-29 (125 records, topic-filtered, 2021–2025), which confirmed 0 new eligible studies after full-text review of 8 title/abstract candidates (5 excluded under E4/E5; 3 excluded under E3 — single-UAV without multi-UAV framework). The corpus remains at 33 studies (see Section 3.1 for full details). Publications in Chinese, Spanish, or Portuguese, or in domain-specific databases (CAB Abstracts, AGRIS), remain outside the search scope. A full Scopus/WoS extension is the highest-priority future action (Section 8.4).
- **Linguistic and geographic bias:** The restriction to English-language publications (criterion I3) introduces a geographic and linguistic bias that warrants explicit acknowledgement. Chinese-language journals publish a substantial volume of research on agricultural UAV systems: institutions such as Northwest A&F University (NWAU), Zhejiang University (ZJU), and China Agricultural University (CAU) are internationally recognised centres for precision agriculture robotics, and their contributions to SI-based multi-UAV path planning may be systematically underrepresented in the present corpus. A preliminary exploratory search of the China National Knowledge Infrastructure (CNKI) using equivalent Chinese-language terms (swarm intelligence, multi-UAV, path planning, precision agriculture) identified approximately 40–60 potentially relevant records published between 2021 and 2025; full screening of this corpus was not feasible within the scope of this review and is reserved as a priority extension for future work (Section 8.4). An equivalent limitation applies to Spanish-language and Portuguese-language publications from Latin American and Iberian research groups active in agricultural UAV research. The English-only restriction may therefore underrepresent SI-based multi-UAV path planning approaches developed specifically for non-Western agricultural contexts—including rice paddy operations, terraced field coverage, and irrigation systems common in Asia and Latin America—that differ structurally from the large-scale cereal field scenarios predominant in the English-language corpus.
- **Single-reviewer title/abstract screening (methodological limitation):** Title and abstract screening of all 481 records was performed by the primary reviewer only, without independent verification by a second reviewer. PRISMA 2020 guidelines [21] recommend that at least two reviewers independently screen titles and abstracts to reduce the risk of systematic exclusion of relevant records. The absence of dual screening at this phase is a methodological limitation of this review: relevant studies may have been excluded at the title/abstract stage without a second reviewer having the opportunity to flag them for full-text assessment. This risk is partially mitigated by three factors: (1) the Boolean search string was validated for 100% recall on a 5-paper gold-standard set; (2) complementary snowballing via Research Rabbit added 20 additional records independently of keyword-based filtering; and (3) the screening threshold at the title/abstract phase was intentionally liberal—borderline records were promoted to full-text assessment rather than excluded—yielding 51 records for full-text review (a conservative approach that reduces the probability of missed relevant studies). Nevertheless, inter-rater agreement at the title/abstract stage cannot be quantified, and the extent of any selection bias introduced at this phase remains unknown. While full-text assessment achieved near-perfect inter-rater agreement ($\kappa = 0.91$; Section 3.2), this does not compensate for the absence of dual screening upstream. This limitation was not corrected prior to submission due to practical constraints and is disclosed here in accordance with PRISMA 2020 transparency requirements (Item 16a). Future reviews in this domain should implement dual independent screening at all phases, with Cohen’s kappa reported for both title/abstract and full-text stages.
- **Inter-rater agreement scope:** While full-text assessment achieved near-perfect agreement ($\kappa = 0.91$), this agreement metric applies only to the 51 full-text records, not to the 481 records screened at title/abstract stage.
- **Gap derivation for studies S20–S33:** 57 of the 171 documented gaps (33.3%) assigned to studies S20–S33 were derived from automated PDF extraction combined with algorithmic pattern matching. Metric-level evidence from the extraction pipeline (`extraer_metricas_v2.py`) confirmed quantitative values for execution time, energy consumption, or convergence criteria in 12 of the 13 affected primary studies. All four gaps previously flagged as “pattern-derived” for study S20 (Ahmed et al., 2021) have been verified against the full-text PDF (DOI: 10.3390/app11083417) and confirmed: simulation-only validation, absence of wind modelling, no scalability analysis beyond four UAVs, and no cross-algorithm SI comparison. No “pattern-derived” assignments remain in the gap taxonomy.
- **MMAT assessments completed:** Two studies (S32, S33) have been verified and classified as High (both High). See Supplementary Table S5 for detailed scores.

• **DRS instrument scope:** The four-criterion DRS instrument (Table 3, Section 3.6) was developed for this review and assesses deployment readiness rather than general methodological quality. It has not been independently validated for psychometric properties. Its criterion weights represent a normative stance (30% weighting of field validation reflects a deliberate value judgment that field evidence is the primary criterion for deployment-relevant research) rather than a neutral quality evaluation. Readers should interpret the resulting Low-DRS classification (18 of 33 studies, 54.5%) in this normative context, and refer to the complementary MMAT results (Section 3.7) for a general methodological quality assessment.

• **Single-institution perspective:** The review was conducted at a single institution, which may introduce unconscious biases in gap prioritisation toward the operational constraints most relevant to the Mexican agricultural context. Expert validation of gap priorities (Supplementary Material S3) partially mitigates this limitation. The authors further disclose that Instituto Tecnológico de Chihuahua II is proposed as the pilot site for AgriSwarm-Bench Phases 1–2; the relationship between the reviewers’ institution and the proposed research agenda is acknowledged as a potential methodological conflict of interest.

8.4. Future Work

Building on the findings and limitations of this review, the following research directions are prioritised for immediate action:

1. **Gap validation for S20 completed:** The four gaps assigned to study S20, previously flagged as pattern-derived, have been verified against the full-text PDF (Ahmed et al., 2021, *Distributed 3-D Path Planning for Multi-UAVs with Full Area Surveillance Based on Particle Swarm Optimization*, DOI: 10.3390/app11083417) and confirmed by direct textual evidence. All gap assignments for studies S20–S33 are now fully verified against primary PDF evidence; no pattern-derived flags remain in the gap taxonomy.
2. **Extend the search** to Scopus and Web of Science, and include non-English publications, to assess whether the algorithm distribution and gap patterns observed in this corpus are robust to a broader search. The supplementary OpenAlex search conducted during revision (125 records, 0 new eligible studies) provides limited evidence of robustness for the single-UAV SI path planning literature; a full Scopus/WoS extension is required to assess multi-UAV and review article coverage.
3. **Extend MMAT to future corpus additions:** As the AgriSwarm-Bench community contributes new benchmark studies, MMAT assessments should be applied systematically using the same instrument (version 2018) and codebook established in this review, enabling longitudinal tracking of methodological quality trends in the SI-based agricultural UAV literature.
4. **Implement AgriSwarm-Bench Phases 1–2** at Instituto Tecnológico de Chihuahua II: develop the S1–S4 benchmark scenarios, establish PSO/ACO/ABC baselines, and complete Stage 2 hardware testbed validation with a 3–5 UAV platform.
5. **Release a public dataset** from the first controlled field experiment on Mendeley Data, providing the agricultural UAV research community with the first open benchmark dataset for SI-based multi-UAV path planning in a real agricultural environment.

Data Availability Statement

The review protocol, search strategies, extraction tools, and supplementary materials are publicly available at the Open Science Framework: <https://doi.org/10.17605/OSF.IO/64DQ9>. The master extraction database, Python scripts, and publication-ready figures will be deposited in Mendeley Data upon manuscript acceptance with a separate DOI for long-term archiving.

Author Contributions

Andres Hidalgo Morales: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Hernán de la Garza Gutiérrez: Supervision, Methodology, Validation (independent study selection, inter-rater agreement), Formal Analysis (Cohen’s kappa calculation), Writing – Review & Editing.

Claudia Georgina Nava Dino: Writing – Review & Editing, Methodological Guidance (PRISMA 2020 compliance), Critical Revision of Gap Taxonomy, Agricultural and Materials Science Domain Expertise, STEM Education Perspective.

Edgar Moisés González Torres: Writing – Review & Editing, Methodological Consultation (aerospace systems design), Critical Feedback on UAV platform constraints and validation protocols.

José Manuel Naveiro Gómez: Writing – Review & Editing, Software Review (computational methods), Validation, Domain Expertise in mechanical engineering simulation and optimization.

Declaration of Competing Interest

The authors declare no financial conflicts of interest. The corresponding author's institutions (Instituto Tecnológico de Chihuahua II and Universidad Politécnica de Chihuahua) are proposed as pilot sites for the AgriSwarm-Bench implementation described in Section 7. This relationship between the authors' institutions and the proposed research agenda is disclosed as a potential methodological conflict of interest.

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List of Abbreviations

ABC	Artificial Bee Colony
ACO	Ant Colony Optimisation
ANFIS	Adaptive Neuro-Fuzzy Inference System
APF	Artificial Potential Field
BVLOS	Beyond Visual Line of Sight
CI	Confidence Interval
COMPAG	Computers and Electronics in Agriculture
CRedit	Contributor Roles Taxonomy
DE	Differential Evolution
DOI	Digital Object Identifier
DRS	Deployment Readiness Score
EES	Elsevier Editorial System
FAA	Federal Aviation Administration (USA)
GA	Genetic Algorithm
GCS	Ground Control Station
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GWO	Grey Wolf Optimizer
ICAO	International Civil Aviation Organization
IEEE	Institute of Electrical and Electronics Engineers

1020 **IMU** Inertial Measurement Unit

1021 **LAI** Leaf Area Index

1022 **LiDAR** Light Detection and Ranging

1023 **MMAT** Mixed Methods Appraisal Tool

1024 **MPC** Model Predictive Control

1025 **NDVI** Normalized Difference Vegetation Index

1026 **OSF** Open Science Framework

1027 **PA** Precision Agriculture

1028 **PICOC** Population, Intervention, Comparison, Outcome, Context

1029 **PRISMA** Preferred Reporting Items for Systematic Reviews and Meta-Analyses

1030 **PSO** Particle Swarm Optimisation

1031 **RPAS** Remotely Piloted Aircraft System (ICAO official term)

1032 **RQ** Research Question

1033 **SI** Swarm Intelligence

1034 **SLAM** Simultaneous Localization and Mapping

1035 **UAS** Unmanned Aircraft System (FAA term; equivalent to RPAS)

1036 **UAV** Unmanned Aerial Vehicle (generic technical term)

1037 **VRA** Variable Rate Application

1038 **WoS** Web of Science

1039 **References**

- 1040 [1] C. Zhang, J. M. Kovacs, The application of small unmanned aerial systems for precision agriculture, Precision Agriculture 13 (4) (2012) 393–408. doi:10.1007/
1041 s11119-012-9274-5.
- 1042 [2] D. C. Tsouros, S. Bibi, P. G. Sarigiannidis, A review on UAV-based applications for precision agriculture, Information 10 (11) (2019) 349. doi:10.3390/
1043 info10110349.
- 1044 [3] V. Puri, A. Nayyar, L. Raja, Agriculture drones: A modern breakthrough in precision agriculture, Journal of Statistics and Management Systems 20 (4) (2017)
1045 507–518. doi:10.1080/09720510.2017.1395171.
- 1046 [4] U. R. Mogili, B. B. V. L. Deepak, Review on application of drone systems in precision agriculture, Procedia Computer Science 133 (2018) 502–509. doi:
1047 10.1016/j.procs.2018.07.063.
- 1048 [5] E. R. Hunt, W. D. Hively, C. S. T. Daughtry, Evaluation of UAV imagery for agricultural crop monitoring, Remote Sensing 10 (4) (2018) 595. doi:10.3390/
1049 rs10040595.
- 1050 [6] S. J. Chung, A. A. Paranjape, P. Dames, S. Shen, V. Kumar, A survey on aerial swarm robotics, IEEE Transactions on Robotics 34 (4) (2018) 837–855. doi:
1051 10.1109/TR0.2018.2857475.
- 1052 [7] L. Gupta, R. Jain, G. Vaszkun, Survey of UAV communication networks, IEEE Communications Surveys and Tutorials 18 (4) (2016) 2595–2621. doi:10.1109/
1053 COMST.2015.2495297.

- [8] J. Huang, Y. Luo, Q. Quan, B. Wang, X. Xue, Y. Zhang, An autonomous task assignment and decision-making method for coverage path planning of multiple pesticide spraying UAVs, *Computers and Electronics in Agriculture* 212 (2023) 108128. doi:10.1016/j.compag.2023.108128.
- [9] I. Bekmezci, O. K. Sahingoz, S. Temel, Flying ad-hoc networks (FANETs): A survey, *Ad Hoc Networks* 11 (3) (2013) 1254–1270. doi:10.1016/j.adhoc.2012.12.004.
- [10] S. Hayat, E. Yanmaz, C. Bettstetter, Experimental analysis of multipoint-to-point UAV communications, *IEEE Communications Letters* 20 (7) (2016) 1433–1436. doi:10.1109/LCOMM.2016.2561274.
- [11] A. Ait Saadi, A. Soukane, Y. Meraihi, A. B. Gabis, S. Mirjalili, A. Ramdane-Cherif, UAV path planning using optimization approaches: A survey, *Archives of Computational Methods in Engineering* 29 (6) (2022) 4233–4284. doi:10.1007/s11831-022-09742-7.
- [12] M. Dorigo, M. Birattari, T. Stutzle, Ant colony optimization, *IEEE Computational Intelligence Magazine* 1 (4) (2006) 28–39. doi:10.1109/MCI.2006.329691.
- [13] J. Kennedy, R. Eberhart, Particle swarm optimization, in: *Proceedings of ICNN'95 – International Conference on Neural Networks*, Vol. 4, 1995, pp. 1942–1948. doi:10.1109/ICNN.1995.488968.
- [14] C. Blum, X. Li, Swarm intelligence in optimization, in: *Swarm Intelligence: Introduction and Applications*, Springer, 2008, pp. 43–85. doi:10.1007/978-3-540-74089-6_2.
- [15] X. S. Yang, *Engineering Optimization: An Introduction with Metaheuristic Applications*, John Wiley & Sons, 2010. doi:10.1002/9780470640425.
- [16] G. Beni, From swarm intelligence to swarm robotics, in: *Swarm Robotics*, Springer, 2005, pp. 1–9. doi:10.1007/978-3-540-30552-1_1.
- [17] E. Bonabeau, M. Dorigo, G. Theraulaz, *Swarm Intelligence: From Natural to Artificial Systems*, Oxford University Press, New York, 1999, ISBN: 978-0-19-513159-8. No DOI assigned (pre-digital monograph, 1999).
- [18] X. S. Yang, M. Karamanoglu, X. He, Flower pollination algorithm: A novel approach for multiobjective optimization, *Engineering Optimization* 46 (9) (2014) 1222–1237, published online 2013; formal print volume year 2014. Cite key yang2013 retained for tex compatibility. doi:10.1080/0305215X.2013.832237.
- [19] L. Li, L. Liu, Y. Shao, X. Zhang, Y. Chen, C. Guo, H. Nian, Enhancing swarm intelligence for obstacle avoidance with multi-strategy and improved dung beetle optimization algorithm, *Sensors* 23 (19) (2023) 8267. doi:10.3390/s23198267.
- [20] C. Xiao, H. Yang, B. Zhang, Multi-unmanned aerial vehicle path planning based on improved nutcracker optimization algorithm, *Drones* 9 (1) (2025) 38. doi:10.3390/drones9010038.
- [21] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, D. Moher, The PRISMA 2020 statement: An updated guideline for reporting systematic reviews, *BMJ* 372 (2021) n71. doi:10.1136/bmj.n71.
- [22] A. Puente-Castro, D. Rivero, A. Pazos, E. Fernandez-Blanco, A review of artificial intelligence applied to path planning in UAV swarms, *Neural Computing and Applications* 34 (1) (2022) 153–170. doi:10.1007/s00521-021-06569-4.
- [23] A. Sharma, S. Shoval, A. Sharma, J. K. Pandey, Path planning for multiple targets interception by the swarm of UAVs based on swarm intelligence algorithms: A review, *IETE Technical Review* 39 (3) (2022) 524–538. doi:10.1080/02564602.2021.1894250.
- [24] M. Brambilla, E. Ferrante, M. Birattari, M. Dorigo, Swarm robotics: A review from the swarm engineering perspective, *Swarm Intelligence* 7 (1) (2013) 1–41. doi:10.1007/s11721-012-0075-2.
- [25] L. Bayindir, A review of swarm robotics tasks, *Neurocomputing* 172 (2016) 292–321. doi:10.1016/j.neucom.2015.05.116.
- [26] M. D. Phung, Q. P. Ha, Safety-enhanced UAV path planning with spherical vector-based particle swarm optimization, *Applied Soft Computing* 107 (2021) 107376. doi:10.1016/j.asoc.2021.107376.
- [27] N. Pan, M. Zhang, Y. Sun, S. Chen, H. Liu, X. Guo, Study on border patrol task planning of heterogeneous UAVs group based on swarm intelligence, *Science Progress* 105 (4) (2022) 00368504221094722, Science Progress uses article IDs in place of page ranges; ID matches DOI suffix. doi:10.1177/00368504221094722.

- [28] M. Shafiq, Z. A. Ali, A. Israr, E. H. Alkhamash, M. Hadjouni, J. J. Jussila, Convergence analysis of path planning of multi-UAVs using max-min ant colony optimization approach, *Sensors* 22 (14) (2022) 5395. doi:10.3390/s22145395.
- [29] Q. Liu, Y. Zhang, M. Li, Z. Zhang, N. Cao, J. Shang, Multi-UAV path planning based on fusion of sparrow search algorithm and improved bioinspired neural network, *IEEE Access* 9 (2021) 120946–120960. doi:10.1109/ACCESS.2021.3109879.
- [30] Q. N. Hong, S. Fàbregues, G. Bartlett, F. Boardman, M. Cargo, P. Dagenais, M.-P. Gagnon, F. Griffiths, B. Nicolau, A. O’Cathain, P. Pluye, The mixed methods appraisal tool (MMAT) version 2018 for information professionals and researchers, *Education for Information* 34 (4) (2018) 285–291. doi:10.3233/EFI-180221.
- [31] J. Chen, F. Ling, Y. Zhang, T. You, Y. Liu, X. Du, Coverage path planning of heterogeneous unmanned aerial vehicles based on ant colony system, *Swarm and Evolutionary Computation* 65 (2021) 101005. doi:10.1016/j.swevo.2021.101005.
- [32] S. Lin, F. Li, X. Li, K. Jia, X. Zhang, Improved artificial bee colony algorithm based on multi-strategy synthesis for UAV path planning, *IEEE Access* 10 (2022) 119269–119285. doi:10.1109/ACCESS.2022.3218685.
- [33] G. Liu, C. Shu, Z. Liang, B. Peng, L. Cheng, A modified sparrow search algorithm with application in 3D route planning for UAV, *Sensors* 21 (4) (2021) 1224. doi:10.3390/s21041224.
- [34] H. Liu, J. Ge, Y. Wang, J. Li, K. Ding, Z. Zhang, Z. Guo, W. Li, J. Lan, Multi-UAV optimal mission assignment and path planning for disaster rescue using adaptive genetic algorithm and improved artificial bee colony method, *Actuators* 11 (1) (2021) 4. doi:10.3390/act11010004.
- [35] A. T. Mathew, A. Paul, A. Rojan, A. Thomas, Implementation of swarm intelligence algorithms for path planning, *Journal of Physics: Conference Series* 1831 (1) (2021) 012008. doi:10.1088/1742-6596/1831/1/012008.
- [36] C. Ntakolia, D. V. Lyridis, A swarm intelligence graph-based pathfinding algorithm based on fuzzy logic (SIGPAF), *Journal of Marine Science and Engineering* 9 (11) (2021) 1243. doi:10.3390/jmse9111243.
- [37] W. Yu, J. Liu, J. Zhou, A novel sparrow particle swarm algorithm (SPSA) for unmanned aerial vehicle path planning, *Scientific Programming* 2021 (2021) 5158304. doi:10.1155/2021/5158304.
- [38] H. Chu, J. Yi, F. Yang, Chaos particle swarm optimization enhancement algorithm for UAV safe path planning, *Applied Sciences* 12 (18) (2022) 8977. doi:10.3390/app12188977.
- [39] G. Fevgas, T. Lagkas, V. Argyriou, P. Sarigiannidis, Coverage path planning methods focusing on energy efficient and cooperative strategies for unmanned aerial vehicles, *Sensors* 22 (3) (2022) 1235. doi:10.3390/s22031235.
- [40] A. Israr, Z. A. Ali, E. H. Alkhamash, J. J. Jussila, Optimization methods applied to motion planning of unmanned aerial vehicles: A review, *Drones* 6 (5) (2022) 126. doi:10.3390/drones6050126.
- [41] Y. Ji, X. Zhao, J. Hao, A novel UAV path planning algorithm based on double-dynamic biogeography-based learning particle swarm optimization, *Mobile Information Systems* 2022 (2022) 8519708. doi:10.1155/2022/8519708.
- [42] B. Selma, S. Chouraqui, B. Selma, H. Abouaissa, Design an optimal ANFIS controller using bee colony optimization for trajectory tracking of a quadrotor UAV, *Journal of The Institution of Engineers (India): Series B* 103 (5) (2022) 1505–1519. doi:10.1007/s40031-022-00747-1.
- [43] N. Ahmed, C. J. Pawase, K. Chang, Distributed 3-D path planning for multi-UAVs with full area surveillance based on particle swarm optimization, *Applied Sciences* 11 (8) (2021) 3417. doi:10.3390/app11083417.
- [44] S. Xu, L. Li, Z. Zhou, Y. Mao, J. Huang, A task allocation strategy of the UAV swarm based on multi-discrete wolf pack algorithm, *Electronics* 11 (3) (2022) 467. doi:10.3390/electronics11030467.
- [45] S. Wang, Y. Zhu, Y. Du, M. Yang, Research on path planning method for mobile platforms based on hybrid swarm intelligence algorithms in multi-dimensional environments, *Applied Sciences* 15 (3) (2025) 1612. doi:10.3390/app15031612.
- [46] L. Deng, H. Chen, X. Zhang, H. Liu, Three-dimensional path planning of UAV based on improved particle swarm optimization, *Drones* 7 (4) (2023) 243. doi:10.3390/drones7040243.
- [47] C. Rao, Z. Wang, P. Shao, A multi-strategy collaborative grey wolf optimization algorithm for UAV path planning, *Drones* 8 (2) (2024) 47. doi:10.3390/drones8020047.

- 1133 [48] W. Zhang, W. Zhang, An efficient UAV localization technique based on particle swarm optimization, *IEEE Transactions on Vehicular Technology* 71 (9) (2022)
1134 9990–10002. doi:10.1109/TVT.2022.3178228.
- 1135 [49] J. Hu, Y. Zheng, S. Wang, C. Zhou, Improved particle swarm optimization based on fuzzy controller fusion of multiple strategies for multi-robot path planning,
1136 *Sensors* 25 (3) (2025) 952. doi:10.3390/s25030952.
- 1137 [50] L. Zuo, Y. Wang, Y. Lu, R. Gu, Hybrid APF–PSO algorithm for regional dynamic formation of UAV swarms, *Drones* 9 (2) (2025) 112. doi:10.3390/
1138 *drones9020112*.
- 1139 [51] X. Yang, S. Zhao, W. Gao, P. Li, Z. Feng, L. Li, T. Jia, X. Wang, Three-dimensional path planning for UAV based on multi-strategy dream optimization algorithm,
1140 *Drones* 9 (1) (2025) 55. doi:10.3390/drones9010055.
- 1141 [52] Y. Yang, X. Xiong, Y. Yan, UAV formation trajectory planning algorithms: A review, *Drones* 7 (1) (2023) 62. doi:10.3390/drones7010062.
- 1142 [53] J. Tang, H. Duan, S. Lao, Swarm intelligence algorithms for multiple unmanned aerial vehicles collaboration: A comprehensive review, *Artificial Intelligence*
1143 *Review* 56 (5) (2022) 4295–4327. doi:10.1007/s10462-022-10281-7.
- 1144 [54] P. Li, L. Wei, D. Wu, An intelligently enhanced ant colony optimization algorithm for global path planning of mobile robots, *Sensors* 25 (5) (2025) 1326. doi:
1145 10.3390/s25051326.
- 1146 [55] Y. Xu, X. Xue, Z. Sun, L. Cui, C. Cai, Q. Zhou, C. Chen, W. Gu, Path planning and scheduling of multiple unmanned ground vehicles for orchard plant protection
1147 operations, *Computers and Electronics in Agriculture*In press (2025). doi:10.1016/j.compag.2025.107215.